CHAPTER 16

Query Optimization

Query optimization is the process of selecting the most efficient query-evaluation plan from among the many strategies usually possible for processing a given query, especially if the query is complex. We do not expect users to write their queries so that they can be processed efficiently. Rather, we expect the system to construct a query-evaluation plan that minimizes the cost of query evaluation. This is where query optimization comes into play.

One aspect of optimization occurs at the relational-algebra level, where the system attempts to find an expression that is equivalent to the given expression, but more efficient to execute. Another aspect is selecting a detailed strategy for processing the query, such as choosing the algorithm to use for executing an operation, choosing the specific indices to use, and so on.

The difference in cost (in terms of evaluation time) between a good strategy and a bad strategy is often substantial and may be several orders of magnitude. Hence, it is worthwhile for the system to spend a substantial amount of time on the selection of a good strategy for processing a query, even if the query is executed only once.

16.1 Overview

Consider the following relational-algebra expression, for the query "Find the names of all instructors in the Music department together with the course title of all the courses that the instructors teach."[1]

$$\Pi_{name,title}\,(\sigma_{dept\_name = \text{"Music"}}\,(instructor \bowtie (teaches \bowtie \Pi_{course\_id,title}(course))))$$

The subexpression $instructor \bowtie teaches \bowtie \Pi_{course\_id,title}(course)$ in the preceding expression can create a very large intermediate result. However, we are interested in only a few tuples of this intermediate result, namely, those pertaining to instructors in the

[1]Note that the projection of *course* on (*course_id*, *title*) is required since *course* shares an attribute *dept_name* with *instructor*; if we did not remove this attribute using the projection, the above expression using natural joins would return only courses from the Music department, even if some Music department instructors taught courses in other departments.

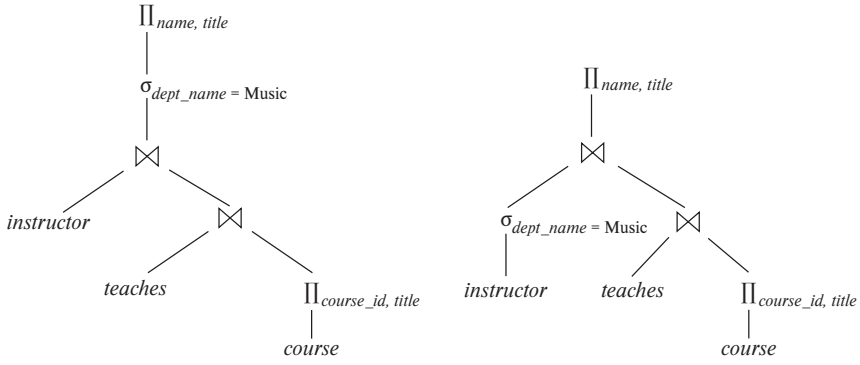


Figure 16.1 Equivalent expressions.

Music department, and in only two of the nine attributes of this relation. Since we are concerned with only those tuples in the *instructor* relation that pertain to the Music department, we do not need to consider those tuples that do not have *dept_name* = "Music". By reducing the number of tuples of the *instructor* relation that we need to access, we reduce the size of the intermediate result. Our query is now represented by the relational-algebra expression:

$$\Pi_{name,title}\left((\sigma_{dept\_name = \text{"Music"}}(instructor)) \bowtie (teaches \bowtie \Pi_{course\_id,title}(course))\right)$$

which is equivalent to our original algebra expression, but which generates smaller intermediate relations. Figure 16.1 depicts the initial and transformed expressions.

An evaluation plan defines exactly what algorithm should be used for each operation and how the execution of the operations should be coordinated. Figure 16.2 illustrates one possible evaluation plan for the expression from Figure 16.1(b). As we have seen, several different algorithms can be used for each relational operation, giving rise to alternative evaluation plans. In the figure, hash join has been chosen for one of the join operations, while the other uses merge join, after sorting the relations on the join attribute, which is ID. All edges are assumed to be pipelined, unless marked as materialized. With pipelined edges the output of the producer is sent directly to the consumer, without being written out to disk; with materialized edges, on the other hand, the output is written to disk, and then read from the disk by the consumer. There are no materialized edges in the evaluation plan in Figure 16.2, although some of the operators, such as sort and hash join, can be represented using suboperators with materialized edges between the suboperators, as we saw in Section 15.7.2.2.

Given a relational-algebra expression, it is the job of the query optimizer to come up with a query-evaluation plan that computes the same result as the given expression, and is the least costly way of generating the result (or, at least, is not much costlier than the least costly way).

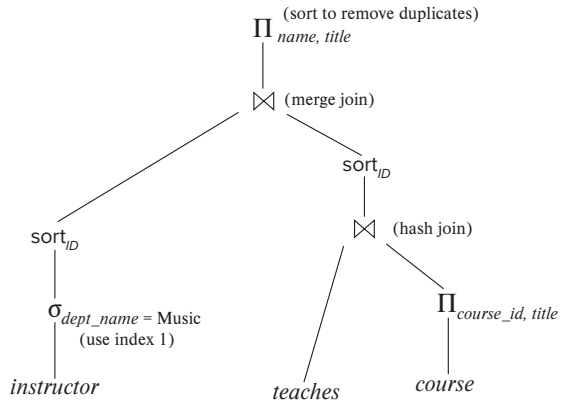


Figure 16.2 An evaluation plan.

The expression that we saw in Figure 16.1 may not necessarily lead to the least-cost evaluation plan for computing the result, since it still computes the join of the entire *teaches* relation with the *course* relation. The following expression gives the same final result, but generates smaller intermediate results, since it joins *teaches* with only *instructor* tuples corresponding to the Music department, and then joins that result with *course*.

$$\Pi_{name,title}\ ((\sigma_{dept\_name = \text{“Music”}}\ (instructor) \bowtie\ teaches) \bowtie\ \Pi_{course\_id,title}(course))$$

Regardless of the way the query is written, it is the job of the optimizer to find the least-cost plan for the query.

To find the least costly query-evaluation plan, the optimizer needs to generate alternative plans that produce the same result as the given expression and to choose the least costly one. Generation of query-evaluation plans involves three steps: (1) generating expressions that are logically equivalent to the given expression, (2) annotating the resultant expressions in alternative ways to generate alternative query-evaluation plans, and (3) estimating the cost of each evaluation plan, and choosing the one whose estimated cost is the least.

Steps (1), (2), and (3) are interleaved in the query optimizer—some expressions are generated and annotated to generate evaluation plans, then further expressions are generated and annotated, and so on. As evaluation plans are generated, their costs are estimated by using statistical information about the relations, such as relation sizes and index depths.

To implement the first step, the query optimizer must generate expressions equivalent to a given expression. It does so by means of *equivalence rules* that specify how to transform an expression into a logically equivalent one. We describe these rules in Section 16.2.

In Section 16.3 we describe how to estimate statistics of the results of each operation in a query plan. Using these statistics with the cost formulae in Chapter 15 allows

Note 16.1 VIEWING QUERY EVALUATION PLANS

Most database systems provide a way to view the evaluation plan chosen to execute a given query. It is usually best to use the GUI provided with the database system to view evaluation plans. However, if you use a command line interface, many databases support variations of a command "**explain** <query>", which displays the execution plan chosen for the specified query <query>. The exact syntax varies with different databases:

- PostgreSQL uses the syntax shown above.
- Oracle uses the syntax **explain plan for**. However, the command stores the resultant plan in a table called *plan_table*, instead of displaying it. The query "**select * from table**(*dbms_xplan.display*);" displays the stored plan.
- DB2 follows a similar approach to Oracle, but requires the program `db2exfmt` to be executed to display the stored plan.
- SQL Server requires the command **set showplan_text on** to be executed before submitting the query; then, when a query is submitted, instead of executing the query, the evaluation plan is displayed.
- MySQL uses the same **explain** <query> syntax as PostgreSQL, but the output is a table whose contents are not easy to understand. However, executing **show warnings** after the **explain** command displays the evaluation plan in a more human-readable format.

The estimated costs for the plan are also displayed along with the plan. It is worth noting that the costs are usually not in any externally meaningful unit, such as seconds or I/O operations, but rather in units of whatever cost model the optimizer uses. Some optimizers such as PostgreSQL display two cost-estimate numbers; the first indicates the estimated cost for outputting the first result, and the second indicates the estimated cost for outputting all results.

us to estimate the costs of individual operations. The individual costs are combined to determine the estimated cost of evaluating a given relational-algebra expression, as outlined in Section 15.7.

In Section 16.4, we describe how to choose a query-evaluation plan. We can choose one based on the estimated cost of the plans. Since the cost is an estimate, the selected plan is not necessarily the least costly plan; however, as long as the estimates are good, the plan is likely to be the least costly one, or not much more costly than it.

Finally, materialized views help to speed up processing of certain queries. In Section 16.5, we study how to "maintain" materialized views—that is, to keep them up-to-date—and how to perform query optimization with materialized views.

16.2 Transformation of Relational Expressions

A query can be expressed in several different ways, with different costs of evaluation. In this section, rather than take the relational expression as given, we consider alternative, equivalent expressions.

Two relational-algebra expressions are said to be **equivalent** if, on every legal database instance, the two expressions generate the same set of tuples. (Recall that a legal database instance is one that satisfies all the integrity constraints specified in the database schema.) Note that the order of the tuples is irrelevant; the two expressions may generate the tuples in different orders, but would be considered equivalent as long as the set of tuples is the same.

In SQL, the inputs and outputs are multisets of tuples, and the multiset version of the relational algebra (described in Note 3.1 on page 80, Note 3.2 on page 97, and Note 3.3 on page 108) is used for evaluating SQL queries. Two expressions in the *multiset* version of the relational algebra are said to be equivalent if on every legal database the two expressions generate the same multiset of tuples. The discussion in this chapter is based on the relational algebra. We leave extensions to the multiset version of the relational algebra to you as exercises.

16.2.1 Equivalence Rules

An **equivalence rule** says that expressions of two forms are equivalent. We can replace an expression of the first form with an expression of the second form, or vice versa—that is, we can replace an expression of the second form by an expression of the first form—since the two expressions generate the same result on any valid database. The optimizer uses equivalence rules to transform expressions into other logically equivalent expressions.

We now describe several equivalence rules on relational-algebra expressions. Some of the equivalences listed appear in Figure 16.3. We use $\theta, \theta_1, \theta_2$, and so on to denote predicates, L_1, L_2, L_3, and so on to denote lists of attributes, and E, E_1, E_2, and so on to denote relational-algebra expressions. A relation name r is simply a special case of a relational-algebra expression and can be used wherever E appears.

1. Conjunctive selection operations can be deconstructed into a sequence of individual selections. This transformation is referred to as a cascade of σ.

$$\sigma_{\theta_1 \wedge \theta_2}(E) \equiv \sigma_{\theta_1}(\sigma_{\theta_2}(E))$$

2. Selection operations are **commutative**.

$$\sigma_{\theta_1}(\sigma_{\theta_2}(E)) \equiv \sigma_{\theta_2}(\sigma_{\theta_1}(E))$$

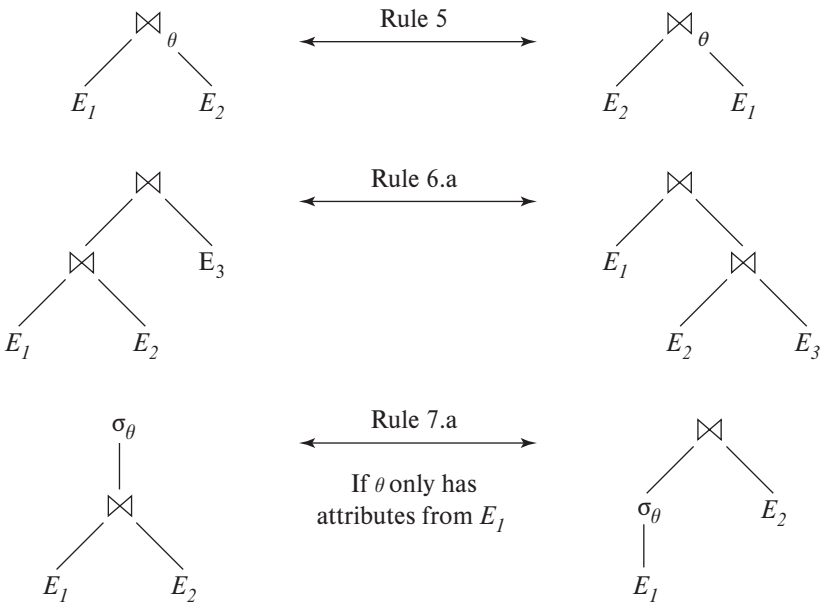


Figure 16.3 Pictorial representation of equivalences.

3. Only the final operations in a sequence of projection operations are needed; the others can be omitted. This transformation can also be referred to as a cascade of Π.

$$\Pi_{L_1}(\Pi_{L_2}(\ldots(\Pi_{L_n}(E))\ldots)) \equiv \Pi_{L_1}(E)$$

where $L_1 \subseteq L_2 \subseteq \ldots \subseteq L_n$.

4. Selections can be combined with Cartesian products and theta joins.

   a. $\sigma_\theta(E_1 \times E_2) \equiv E_1 \bowtie_\theta E_2$

      This expression is just the definition of the theta join.

   b. $\sigma_{\theta_1}(E_1 \bowtie_{\theta_2} E_2) \equiv E_1 \bowtie_{\theta_1 \wedge \theta_2} E_2$

5. Theta-join operations are commutative.

$$E_1 \bowtie_\theta E_2 \equiv E_2 \bowtie_\theta E_1$$

Recall that the natural-join operator is simply a special case of the theta-join operator; hence, natural joins are also commutative.

The order of attributes differs between the left-hand side and right-hand side of the commutativity rule, so the equivalence does not hold if the order of attributes is taken into account. Since we use a version of relational algebra where every attribute must have a name for it to be referenced, the order of attributes

does not actually matter, except when the result is finally displayed. When the order does matter, a projection operation can be added to one of the sides of the equivalence to appropriately reorder attributes. However, for simplicity, we omit the projection and ignore the attribute order in all our equivalence rules.

6. a. Natural-join operations are **associative**.

$$(E_1 \bowtie E_2) \bowtie E_3 \equiv E_1 \bowtie (E_2 \bowtie E_3)$$

b. Theta joins are associative in the following manner:

$$(E_1 \bowtie_{\theta_1} E_2) \bowtie_{\theta_2 \wedge \theta_3} E_3 \equiv E_1 \bowtie_{\theta_1 \wedge \theta_3} (E_2 \bowtie_{\theta_2} E_3)$$

where θ_2 involves attributes from only E_2 and E_3. Any of these conditions may be empty; hence, it follows that the Cartesian product ($\times$) operation is also associative. The commutativity and associativity of join operations are important for join reordering in query optimization.

7. The selection operation distributes over the theta-join operation under the following two conditions:

a. Selection distributes over the theta-join operation when all the attributes in selection condition θ_1 involve only the attributes of one of the expressions (say, E_1) being joined.

$$\sigma_{\theta_1}(E_1 \bowtie_{\theta} E_2) \equiv (\sigma_{\theta_1}(E_1)) \bowtie_{\theta} E_2$$

b. Selection distributes over the theta-join operation when selection condition θ_1 involves only the attributes of E_1 and θ_2 involves only the attributes of E_2.

$$\sigma_{\theta_1 \wedge \theta_2}(E_1 \bowtie_{\theta} E_2) \equiv (\sigma_{\theta_1}(E_1)) \bowtie_{\theta} (\sigma_{\theta_2}(E_2))$$

8. The projection operation distributes over the theta-join operation under the following conditions.

a. Let L_1 and L_2 be attributes of E_1 and E_2, respectively. Suppose that the join condition θ involves only attributes in $L_1 \cup L_2$. Then,

$$\Pi_{L_1 \cup L_2}(E_1 \bowtie_{\theta} E_2) \equiv (\Pi_{L_1}(E_1)) \bowtie_{\theta} (\Pi_{L_2}(E_2))$$

b. Consider a join $E_1 \bowtie_{\theta} E_2$. Let L_1 and L_2 be sets of attributes from E_1 and E_2, respectively. Let L_3 be attributes of E_1 that are involved in join condition θ, but are not in L_1 and let L_4 be attributes of E_2 that are involved in join condition θ, but are not in L_2. Then,

$$\Pi_{L_1 \cup L_2}(E_1 \bowtie_{\theta} E_2) \equiv \Pi_{L_1 \cup L_2}((\Pi_{L_1 \cup L_3}(E_1)) \bowtie_{\theta} (\Pi_{L_2 \cup L_4}(E_2)))$$

Similar equivalences hold for outer join operations $\mathbin{⟕}$, $\mathbin{⟗}$ and $\mathbin{⟖}$.

9. The set operations union and intersection are commutative.

   a. $E_1 \cup E_2 \equiv E_2 \cup E_1$

   b. $E_1 \cap E_2 \equiv E_2 \cap E_1$

   Set difference is not commutative.

10. Set union and intersection are associative.

    a. $(E_1 \cup E_2) \cup E_3 \equiv E_1 \cup (E_2 \cup E_3)$

    b. $(E_1 \cap E_2) \cap E_3 \equiv E_1 \cap (E_2 \cap E_3)$

11. The selection operation distributes over the union, intersection, and set-difference operations.

    a. $\sigma_\theta(E_1 \cup E_2) \equiv \sigma_\theta(E_1) \cup \sigma_\theta(E_2)$

    b. $\sigma_\theta(E_1 \cap E_2) \equiv \sigma_\theta(E_1) \cap \sigma_\theta(E_2)$

    c. $\sigma_\theta(E_1 - E_2) \equiv \sigma_\theta(E_1) - \sigma_\theta(E_2)$

    d. $\sigma_\theta(E_1 \cap E_2) \equiv \sigma_\theta(E_1) \cap E_2$

    e. $\sigma_\theta(E_1 - E_2) \equiv \sigma_\theta(E_1) - E_2$

    The preceding equivalence does not hold if $-$ is replaced by $\cup$.

12. The projection operation distributes over the union operation

$$\Pi_L(E_1 \cup E_2) \equiv (\Pi_L(E_1)) \cup (\Pi_L(E_2))$$

    provided E_1 and E_2 have the same schema.

13. Selection distributes over aggregation under the following conditions. Let G be a set of group by attributes, and A a set of aggregate expressions. When θ only involves attributes in G, the following equivalence holds.

$$\sigma_\theta({}_G\gamma_A(E) \equiv {}_G\gamma_A(\sigma_\theta(E))$$

14. a. Full outer join is commutative.

$$E_1 ⟗ E_2 \equiv E_2 ⟗ E_2$$

    b. Left and right outer join are not commutative. However, left outer join and right outer join can be exchanged as follows.

$$E_1 ⟕ E_2 \equiv E_2 ⟖ E_1$$

15. Selection distributes over left and right outer join under some conditions. Specifically, when the selection condition θ_1 involves only the attributes of one of the expressions being joined, say E_1, the following equivalences hold.

a. $\sigma_{\theta_1}(E_1 \rtimes_\theta E_2) \equiv (\sigma_{\theta_1}(E_1)) \rtimes_\theta E_2$

b. $\sigma_{\theta_1}(E_2 \ltimes_\theta E_1) \equiv (E_2 \ltimes_\theta (\sigma_{\theta_1}(E_1)))$

16. Outer joins can be replaced by inner joins under some conditions. Specifically, if θ_1 has the property that it evaluates to false or unknown whenever the attributes of E_2 are null, then the following equivalences hold.

a. $\sigma_{\theta_1}(E_1 \rtimes_\theta E_2) \equiv \sigma_{\theta_1}(E_1 \bowtie_\theta E_2)$

b. $\sigma_{\theta_1}(E_2 \ltimes_\theta E_1) \equiv \sigma_{\theta_1}(E_2 \bowtie_\theta E_2)$

A predicate θ_1 satisfying the above property is said to be **null rejecting** on E_2. For example, if θ_1 is of the form $A < 4$ where A is an attribute from E_2, then θ_1 would evaluate to unknown whenever A is null, and as a result any tuples in $E_1 \rtimes_\theta E_2$ that are not in $E_1 \bowtie_\theta E_2$ would be rejected by σ_{θ_1}. We can therefore replace the outer join by an inner join (or vice versa).

More generally, the condition would hold if θ_1 is of the form $\theta_1^1 \wedge \theta_1^2 \wedge \ldots \wedge \theta_1^k$, and at least one of the terms θ_1^i is of the form e_1 *relop* e_2, where e_1 and e_2 are arithmetic or string expressions involving at least one attribute from E_2, and *relop* is any of $<, \leq, =, \geq, >$.

This is only a partial list of equivalences. More equivalences are discussed in the exercises.

Some equivalences that hold for joins do not hold for outer joins. For example, the selection operation does not distribute over outer join when the conditions specified in rule 15.a or rule 15.b do hold. To see this, we look at the expression:

$$\sigma_{year=2017}(instructor \rtimes teaches)$$

and consider the case of an instructor who teaches no courses at all, regardless of year. In the above expression, the left outer join retains a tuple for each such instructor with a null value for *year*. Then the selection operation removes those tuples since the predicate *null*=2017 evaluates to *unknown*, and such instructors do not appear in the result. However, if we push the selection operation down to *teaches*, the resulting expression:

$$instructor \rtimes \sigma_{year=2017}(teaches)$$

is syntactically correct since the selection predicate includes only attributes from *teaches*, but the result is different. For an instructor that does not teach at all, the *instructor* tuple appears in the result of $instructor \rtimes \sigma_{year=2017}(teaches)$, but not in the result of $\sigma_{year=2017}(instructor \rtimes teaches)$. The following equivalence, however, does hold:

$$\sigma_{year=2017}(instructor \rtimes teaches) \equiv \sigma_{year=2017}(instructor \bowtie teaches)$$

As another example, unlike inner joins, outer joins are not associative. We show thus using an example for the natural left outer join. Similar examples can be con-

structed for natural right and natural full outer join, as well as for the corresponding theta-join versions of the outer join operations.

Let relation $r(A, B)$ be a relation consisting of the single tuple $(1, 1)$, $s(B, C)$ be a relation consisting of the single tuple $(1, 1)$, and $t(A, C)$ be an empty relation with no tuples. We shall show that for this example,

$$(r \rtimes s) \rtimes t \not\equiv r \rtimes (s \rtimes t)$$

To see this, note first that $(r \rtimes s)$ produces a result with schema (A, B, C) having one tuple $(1, 1, 1)$. Computing the left outer join of that result with relation t produces a result with schema (A, B, C) having one tuple $(1, 1, 1)$. Next, we look at the expression $r \rtimes (s \rtimes t)$, and note that $s \rtimes t$ produces a result with schema (A, B, C) having one tuple $(null, 1, 1)$. Computing the left outer join of r with that result produces a result with schema (A, B, C) having one tuple $(1, 1, null)$.

16.2.2 Examples of Transformations

We now illustrate the use of the equivalence rules. We use our university example with the relation schemas:

instructor(*ID*, *name*, *dept_name*, *salary*)
teaches(*ID*, *course_id*, *sec_id*, *semester*, *year*)
course(*course_id*, *title*, *dept_name*, *credits*)

In our example in Section 16.1, the expression:

$$\Pi_{name,title}\,(\sigma_{dept\_name = \text{“Music”}}\,(instructor \bowtie (teaches \bowtie \Pi_{course\_id,title}(course))))$$

was transformed into the following expression:

$$\Pi_{name,title}\,((\sigma_{dept\_name = \text{“Music”}}\,(instructor)) \bowtie (teaches \bowtie \Pi_{course\_id,title}(course)))$$

which is equivalent to our original algebra expression but generates smaller intermediate relations. We can carry out this transformation by using rule 7.a. Remember that the rule merely says that the two expressions are equivalent; it does not say that one is better than the other.

Multiple equivalence rules can be used, one after the other, on a query or on parts of the query. As an illustration, suppose that we modify our original query to restrict attention to instructors who have taught a course in 2017. The new relational-algebra query is:

$$\Pi_{name,title}\,(\sigma_{dept\_name = \text{“Music”} \wedge year\ =\ 2017}\,(instructor \bowtie (teaches \bowtie \Pi_{course\_id,title}(course))))$$

We cannot apply the selection predicate directly to the *instructor* relation, since the predicate involves attributes of both the *instructor* and *teaches* relations. However, we can first apply rule 6.a (associativity of natural join) to transform the join $instructor \bowtie (teaches \bowtie \Pi_{course\_id,title}(course))$ into $(instructor \bowtie teaches) \bowtie \Pi_{course\_id,title}(course)$:

$$\Pi_{name,title}\ (\sigma_{dept\_name = \text{“Music”} \wedge year\ =\ 2017} ((instructor \bowtie teaches) \bowtie \Pi_{course\_id,title}(course)))$$

Then, using rule 7.a, we can rewrite our query as:

$$\Pi_{name,title}\ ((\sigma_{dept\_name = \text{“Music”} \wedge year\ =\ 2017} (instructor \bowtie teaches)) \bowtie \Pi_{course\_id,title}(course))$$

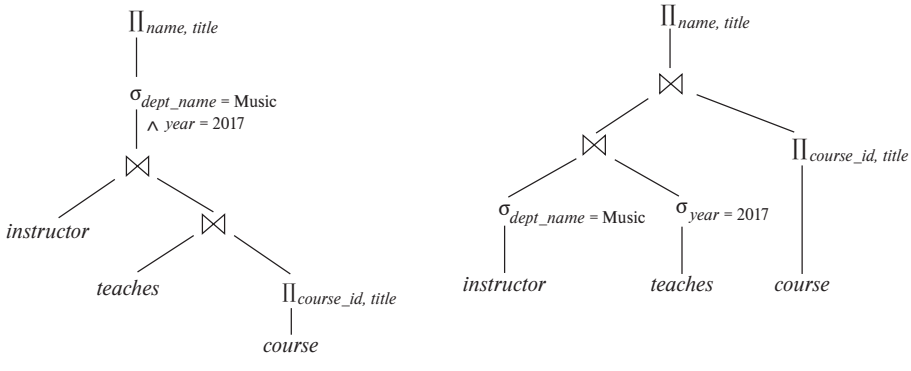
Let us examine the selection subexpression within this expression. Using rule 1, we can break the selection into two selections to get the following subexpression:

$$\sigma_{dept\_name = \text{“Music”}}\ (\sigma_{year\ =\ 2017}\ (instructor \bowtie teaches))$$

Both of the preceding expressions select tuples with *dept_name* = “Music” and *course_id* = 2017. However, the latter form of the expression provides a new opportunity to apply rule 7.a (“perform selections early”), resulting in the subexpression:

$$\sigma_{dept\_name = \text{“Music”}}\ (instructor) \bowtie \sigma_{year\ =\ 2017}\ (teaches)$$

Figure 16.4 depicts the initial expression and the final expression after all these transformations. We could equally well have used rule 7.b to get the final expression directly, without using rule 1 to break the selection into two selections. In fact, rule 7.b can itself be derived from rules 1 and 7.a.



(a) Initial expression tree

(b) Tree after multiple transformations

Figure 16.4 Multiple transformations.

A set of equivalence rules is said to be **minimal** if no rule can be derived from any combination of the others. The preceding example illustrates that the set of equivalence rules in Section 16.2.1 is not minimal. An expression equivalent to the original expression may be generated in different ways; the number of different ways of generating an expression increases when we use a nonminimal set of equivalence rules. Query optimizers therefore use minimal sets of equivalence rules.

Now consider the following form of our example query:

$$\Pi_{name,title}\ ((\sigma_{dept\_name = \text{“Music”}}\ (instructor) \bowtie teaches) \bowtie \Pi_{course\_id,title}(course))$$

When we compute the subexpression:

$$(\sigma_{dept\_name = \text{“Music”}}\ (instructor) \bowtie teaches)$$

we obtain a relation whose schema is:

$$(ID, name, dept\_name, salary, course\_id, sec\_id, semester, year)$$

We can eliminate several attributes from the schema by pushing projections based on equivalence rules 8.a and 8.b. The only attributes that we must retain are those that either appear in the result of the query or are needed to process subsequent operations. By eliminating unneeded attributes, we reduce the number of columns of the intermediate result. Thus, we reduce the size of the intermediate result. In our example, the only attributes we need from the join of *instructor* and *teaches* are *name* and *course_id*. Therefore, we can modify the expression to:

$$\Pi_{name,title}\ ((\Pi_{name,course\_id}\ ((\sigma_{dept\_name = \text{“Music”}}\ (instructor)) \bowtie teaches)) \bowtie \Pi_{course\_id,title}(course))$$

The projection $\Pi_{name,course\_id}$ reduces the size of the intermediate join results.

16.2.3 Join Ordering

A good ordering of join operations is important for reducing the size of temporary results; hence, most query optimizers pay a lot of attention to the join order. As mentioned in equivalence rule 6.a, the natural-join operation is associative. Thus, for all relations r_1, r_2, and r_3:

$$(r_1 \bowtie r_2) \bowtie r_3 \equiv r_1 \bowtie (r_2 \bowtie r_3)$$

Although these expressions are equivalent, the costs of computing them may differ. Consider again the expression:

$$\Pi_{name,title}\ ((\sigma_{dept\_name = \text{“Music”}}\ (instructor)) \bowtie teaches \bowtie \Pi_{course\_id,title}(course))$$

We could choose to compute $teaches \bowtie \Pi_{course\_id,title}(course)$ first, and then to join the result with:

$$\sigma_{dept\_name = \text{“Music”}}(instructor)$$

However, $teaches \bowtie \Pi_{course\_id,title}(course)$ is likely to be a large relation, since it contains one tuple for every course taught. In contrast:

$$\sigma_{dept\_name = \text{“Music”}}(instructor) \bowtie teaches$$

is probably a small relation. To see that it is, we note that a university has fewer instructors than courses and, since a university has a large number of departments, it is likely that only a small fraction of the university instructors are associated with the Music department. Thus, the preceding expression results in one tuple for each course taught by an instructor in the Music department. Therefore, the temporary relation that we must store is smaller than it would have been had we computed $teaches \bowtie \Pi_{course\_id,title}(course)$ first.

There are other options to consider for evaluating our query. We do not care about the order in which attributes appear in a join, since it is easy to change the order before displaying the result. Thus, for all relations r_1 and r_2:

$$r_1 \bowtie r_2 \equiv r_2 \bowtie r_1$$

That is, natural join is commutative (equivalence rule 5).

Using the associativity and commutativity of the natural join (rules 5 and 6), consider the following relational-algebra expression:

$$(instructor \bowtie \Pi_{course\_id,title}(course)) \bowtie teaches$$

Note that there are no attributes in common between $\Pi_{course\_id,title}(course)$ and *instructor*, so the join is just a Cartesian product. If there are a tuples in *instructor* and b tuples in $\Pi_{course\_id,title}(course)$, this Cartesian product generates $a * b$ tuples, one for every possible pair of instructor tuple and course (without regard for whether the instructor taught the course). This Cartesian product would produce a very large temporary relation. However, if the user had entered the preceding expression, we could use the associativity and commutativity of the natural join to transform this expression to the more efficient expression:

$$(instructor \bowtie teaches) \bowtie \Pi_{course\_id,title}(course)$$

16.2.4 Enumeration of Equivalent Expressions

Query optimizers can use equivalence rules to systematically generate expressions equivalent to the given query expression. The cost of an expression is computed based

```
procedure genAllEquivalent(E)
EQ = {E}
repeat
    Match each expression E_i in EQ with each equivalence rule R_j
    if any subexpression e_i of E_i matches one side of R_j
        Create a new expression E' which is identical to E_i, except that
            e_i is transformed to match the other side of R_j
        Add E' to EQ if it is not already present in EQ
until no new expression can be added to EQ
```

Figure 16.5 Procedure to generate all equivalent expressions.

on statistics that are discussed in Section 16.3. Cost-based query optimizers, described in Section 16.4 compute the cost of each alternative and pick the least cost alternative.

Conceptually, enumeration of equivalent expressions can be done as outlined in Figure 16.5. The process proceeds as follows: Given a query expression E, the set of equivalent expressions EQ initially contains only E. Now, each expression in EQ is matched with each equivalence rule. If a subexpression e_j of any expression $E_i \in EQ$ (as a special case, e_j could be E_i itself) matches one side of an equivalence rule, the optimizer generates a copy E_k of E_i, in which e_j is transformed to match the other side of the rule, and adds E_k to EQ. This process continues until no more new expressions can be generated. With a properly chosen set of equivalence rules, the set of equivalent expressions is finite, and the process can be guaranteed to terminate.

For example, given an expression $r \bowtie (s \bowtie t)$, the commutativity rule can match the subexpression $(s \bowtie t)$, and would create a new expression $r \bowtie (t \bowtie s)$. The commutativity rule also matches the join at the root of $r \bowtie (s \bowtie t)$, and creates a new expression $(s \bowtie t) \bowtie r$. Associativity and commutativity rules can continue to be applied to generate new expressions. But eventually applying any equivalence rule will only generate expressions that were already generated earlier, and the process will terminate.

The preceding process is extremely costly both in space and in time, but optimizers can greatly reduce both the space and time cost, using two key ideas.

1. If we generate an expression E' from an expression E_1 by using an equivalence rule on subexpression e_i, then E' and E_1 have identical subexpressions except for e_i and its transformation. Even e_i and its transformed version usually share many identical subexpressions. Expression-representation techniques that allow both expressions to point to shared subexpressions can reduce the space requirement significantly.

2. It is not always necessary to generate every expression that can be generated with the equivalence rules. If an optimizer takes cost estimates of evaluation into account, it may be able to avoid examining some of the expressions, as we shall see in Section 16.4. We can reduce the time required for optimization by using techniques such as these.

With these and other techniques to reduce the optimization time, equivalence rules can be used to enumerate alternative plans, whose costs can be computed; the lowest-cost plan amongst the alternatives is then chosen. We discuss efficient implementation of cost-based query optimization based on equivalence rules in Section 16.4.2.

Some query optimizers use equivalence rules in a heuristic manner. With such an approach, if the left-hand side of an equivalence rule matches a subtree in a query plan, the subtree is rewritten to match the right-hand side of the rule. This process is repeated till the query plan cannot be further rewritten. Rules must be carefully chosen such that the cost decreases when a rule is applied, and rewriting must eventually terminate. Although this approach can be implemented to execute quite fast, there is no guarantee that it will find the optimal plan.

Yet other query optimizers focus on join order selection, which is often a key factor in query cost. We discuss algorithms for join-order optimization in Section 16.4.1.

16.3 Estimating Statistics of Expression Results

The cost of an operation depends on the size and other statistics of its inputs. Given an expression such as $r \bowtie (s \bowtie t)$ to estimate the cost of joining r with $(s \bowtie t)$, we need to have estimates of statistics such as the size of $s \bowtie t$.

In this section, we first list some statistics about database relations that are stored in database-system catalogs, and then show how to use the stored statistics to estimate statistics on the results of various relational operations.

Given a query expression, we consider it as a tree; we can start from the bottom-level operations, and estimate their statistics, and continue the process on higher-level operations, till we reach the root of the tree. The size estimates that we compute as part of these statistics can be used to compute the cost of algorithms for individual operations in the tree, and these costs can be added up to find the cost of an entire query plan, as we saw in Chapter 15.

One thing that will become clear later in this section is that the estimates are not very accurate, since they are based on assumptions that may not hold exactly. A query-evaluation plan that has the lowest estimated execution cost may therefore not actually have the lowest actual execution cost. However, real-world experience has shown that even if estimates are not precise, the plans with the lowest estimated costs usually have actual execution costs that are either the lowest actual execution costs or are close to the lowest actual execution costs.

16.3.1 Catalog Information

The database-system catalog stores the following statistical information about database relations:

- n_r, the number of tuples in the relation r.
- b_r, the number of blocks containing tuples of relation r.
- l_r, the size of a tuple of relation r in bytes.
- f_r, the blocking factor of relation r—that is, the number of tuples of relation r that fit into one block.
- $V(A, r)$, the number of distinct values that appear in the relation r for attribute A. This value is the same as the size of $\Pi_A(r)$. If A is a key for relation r, $V(A, r)$ is n_r.

The last statistic, $V(A, r)$, can also be maintained for sets of attributes, if desired, instead of just for individual attributes. Thus, given a set of attributes, $\mathcal{A}$, $V(\mathcal{A}, r)$ is the size of $\Pi_{\mathcal{A}}(r)$.

If we assume that the tuples of relation r are stored together physically in a file, the following equation holds:

$$b_r = \left\lceil \frac{n_r}{f_r} \right\rceil$$

Statistics about indices, such as the heights of B^+-tree indices and number of leaf pages in the indices, are also maintained in the catalog.

If we wish to maintain accurate statistics, then every time a relation is modified, we must also update the statistics. This update incurs a substantial amount of overhead. Therefore, most systems do not update the statistics on every modification. Instead, they update the statistics during periods of light system load. As a result, the statistics used for choosing a query-processing strategy may not be completely accurate. However, if not too many updates occur in the intervals between the updates of the statistics, the statistics will be sufficiently accurate to provide a good estimation of the relative costs of the different plans.

The statistical information noted here is simplified. Real-world optimizers often maintain further statistical information to improve the accuracy of their cost estimates of evaluation plans. For instance, most databases store the distribution of values for each attribute as a **histogram**: in a histogram, the values for the attribute are divided into a number of ranges, and with each range the histogram associates the number of tuples whose attribute value lies in that range. Figure 16.6 shows an example of a histogram for an integer-valued attribute that takes values in the range 1 to 25.

As an example of a histogram, the range of values for an attribute *age* of a relation *person* could be divided into 0–9, 10–19, . . . , 90–99 (assuming a maximum age of 99). With each range we store a count of the number of *person* tuples whose *age* values lie in that range.

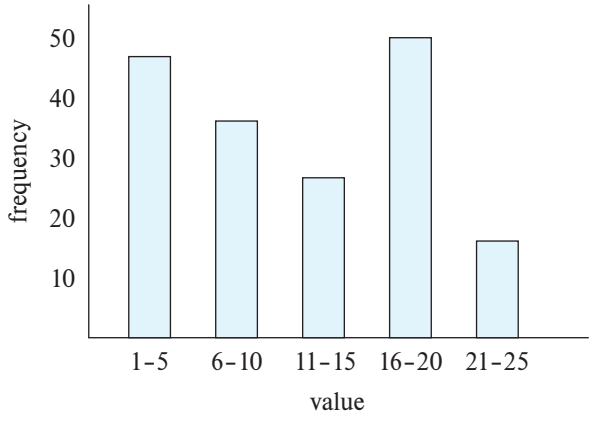


Figure 16.6 Example of histogram.

The histogram shown in Figure 16.6, is an **equi-width histogram** since it divides the range of values into equal-sized ranges. In contrast, an **equi-depth** histogram adjusts the boundaries of the ranges such that each range has the same number of values. Thus, an equi-depth histogram merely stores the boundaries of partitions of the range, and need not store the number of values. For example, the following could be the equidepth histogram for the data whose equi-width histogram is shown in Figure 16.6:

(4, 8, 14, 19)

The histogram indicates that 1/5th of the tuples have age less than 4, another 1/5th have age ≥ 4 but < 8, and so on, with the last 1/5th having age ≥ 19. Information about the total number of tuples is also stored with the equi-width histogram. Equi-depth histograms are preferred to equi-width histograms since they provide better estimates, and occupy less space.

Histograms used in database systems can also record the number of distinct values in each range, in addition to the number of tuples with attribute values in that range. In our example, the histogram could store the number of distinct age values that lie in each range. Without such histogram information, an optimizer would have to assume that the distribution of values is uniform; that is, each range has the same number of distinct values.

In many database applications, some values are very frequent, compared to other values. To get better estimates for queries that specify these values, many databases store a list of n most frequent values for some n (say 5 or 10), along with the number of times each value appears. In our example, if ages 4, 7, 18, 19, and 23 are the five most frequently occurring values, the database could store the number of persons having each of these ages. The histogram then only stores statistics for age values other than these five values, since we have now have exact counts for these values.

A histogram takes up only a little space, so histograms on several different attributes can be stored in the system catalog.

16.3.2 Selection Size Estimation

The size estimate of the result of a selection operation depends on the selection predicate. We first consider a single equality predicate, then a single comparison predicate, and finally combinations of predicates.

- $\sigma_{A=a}(r)$: If a is a frequently occurring value for which the occurrence count is available, we can use that value directly as the size estimate for the selection.

  Otherwise if there is no histogram available, we assume uniform distribution of values (i.e., each value appears with equal probability), the selection result is estimated to have $n_r/V(A,r)$ tuples, assuming that the value a appears in attribute A of some record of r. The assumption that the value a in the selection appears in some record is generally true, and cost estimates often make it implicitly. However, it is often not realistic to assume that each value appears with equal probability. The *course_id* attribute in the *takes* relation is an example where the assumption is not valid. It is reasonable to expect that a popular undergraduate course will have many more students than a smaller specialized graduate course. Therefore, certain *course_id* values appear with greater probability than do others. Despite the fact that the uniform-distribution assumption is often not correct, it is a reasonable approximation of reality in many cases, and it helps us to keep our presentation relatively simple.

  If a histogram is available on attribute A, we can locate the range that contains the value a, and modify the above-mentioned estimate $n_r/V(A,r)$ by using the frequency count for that range instead of n_r, and the number of distinct values that occurs in that range instead of $V(A,r)$.

- $\sigma_{A\leq v}(r)$: Consider a selection of the form $\sigma_{A\leq v}(r)$. Suppose that the lowest and highest values ($\min(A,r)$ and $\max(A,r)$) for the attribute are stored in the catalog. Assuming that values are uniformly distributed, we can estimate the number of records that will satisfy the condition $A \leq v$ as:

  - 0 if $v < \min(A,r)$
  - n_r if $v \geq \max(A,r)$, and,
  - $n_r \cdot \frac{v-\min(A,r)}{\max(A,r)-\min(A,r)}$, otherwise.

  If a histogram is available on attribute A, we can get a more accurate estimate; we leave the details as an exercise for you.

  In some cases, such as when the query is part of a stored procedure, the value v may not be available when the query is optimized. In such cases, we assume that approximately one-half the records will satisfy the comparison condition. That is,

Note 16.2 COMPUTING AND MAINTAINING STATISTICS

Conceptually, statistics on relations can be thought of as materialized views, which should be automatically maintained when relations are modified. Unfortunately, keeping statistics up-to-date on every insert, delete or update to the database can be very expensive. On the other hand, optimizers generally do not need exact statistics: an error of a few percent may result in a plan that is not quite optimal being chosen, but the alternative plan chosen is likely to have a cost which is within a few percent of the optimal cost. Thus, it is acceptable to have statistics that are approximate.

Database systems reduce the cost of generating and maintaining statistics, as outlined below, by exploiting the fact that statistics can be approximate.

- Statistics are often computed from a sample of the underlying data, instead of examining the entire collection of data. For example, a fairly accurate histogram can be computed from a sample of a few thousand tuples, even on a relation that has millions, or hundreds of millions of records. However, the sample used must be a **random sample**; a sample that is not random may have an excessive representation of one part of the relation and can give misleading results. For example, if we used a sample of instructors to compute a histogram on salaries, if the sample has an overrepresentation of lower-paid instructors the histogram would result in wrong estimates. Database systems today routinely use random sampling to create statistics. See the bibliographical notes online for references on sampling.
- Statistics are not maintained on every update to the database. In fact, some database systems never update statistics automatically. They rely on database administrators periodically running a command to update statistics. Oracle and PostgreSQL provide an SQL command called **analyze** that generates statistics on specified relations, or on all relations. IBM DB2 supports an equivalent command called **runstats**. See the system manuals for details. You should be aware that optimizers sometimes choose very bad plans due to incorrect statistics. Many database systems, such as IBM DB2, Oracle, and SQL Server, update statistics automatically at certain points of time. For example, the system can keep approximate track of how many tuples there are in a relation and recompute statistics if this number changes significantly. Another approach is to compare estimated cardinalities of a relation scan with actual cardinalities when a query is executed, and if they differ significantly, initiate an update of statistics for that relation.

we assume the result has $n_r/2$ tuples; the estimate may be very inaccurate, but it is the best we can do without any further information.

- Complex selections:
  - **Conjunction:** A *conjunctive selection* is a selection of the form:

$$\sigma_{\theta_1 \wedge \theta_2 \wedge \cdots \wedge \theta_n}(r)$$

We can estimate the result size of such a selection: For each θ_i, we estimate the size of the selection $\sigma_{\theta_i}(r)$, denoted by s_i, as described previously. Thus, the probability that a tuple in the relation satisfies selection condition θ_i is s_i/n_r.

The preceding probability is called the **selectivity** of the selection $\sigma_{\theta_i}(r)$. Assuming that the conditions are *independent* of each other, the probability that a tuple satisfies all the conditions is simply the product of all these probabilities. Thus, we estimate the number of tuples in the full selection as:

$$n_r * \frac{s_1 * s_2 * \cdots * s_n}{n_r^n}$$

  - **Disjunction:** A *disjunctive selection* is a selection of the form:

$$\sigma_{\theta_1 \vee \theta_2 \vee \cdots \vee \theta_n}(r)$$

A disjunctive condition is satisfied by the union of all records satisfying the individual, simple conditions θ_i.

As before, let s_i/n_r denote the probability that a tuple satisfies condition θ_i. The probability that the tuple will satisfy the disjunction is then 1 minus the probability that it will satisfy *none* of the conditions:

$$1 - (1 - \frac{s_1}{n_r}) * (1 - \frac{s_2}{n_r}) * \cdots * (1 - \frac{s_n}{n_r})$$

Multiplying this value by n_r gives us the estimated number of tuples that satisfy the selection.

  - **Negation:** In the absence of nulls, the result of a selection $\sigma_{\neg\theta}(r)$ is simply the tuples of r that are not in $\sigma_\theta(r)$. We already know how to estimate the number of tuples in $\sigma_\theta(r)$. The number of tuples in $\sigma_{\neg\theta}(r)$ is therefore estimated to be n_r minus the estimated number of tuples in $\sigma_\theta(r)$.

We can account for nulls by estimating the number of tuples for which the condition θ would evaluate to *unknown*, and subtracting that number from the above estimate, ignoring nulls. Estimating that number would require extra statistics to be maintained in the catalog.

16.3.3 Join Size Estimation

In this section, we see how to estimate the size of the result of a join.

The Cartesian product $r \times s$ contains $n_r * n_s$ tuples. Each tuple of $r \times s$ occupies $l_r + l_s$ bytes, from which we can calculate the size of the Cartesian product.

Estimating the size of a natural join is somewhat more complicated than estimating the size of a selection or of a Cartesian product. Let $r(R)$ and $s(S)$ be relations.

- If $R \cap S = \emptyset$—that is, the relations have no attribute in common—then $r \bowtie s$ is the same as $r \times s$, and we can use our estimation technique for Cartesian products.
- If $R \cap S$ is a key for R, then we know that a tuple of s will join with at most one tuple from r. Therefore, the number of tuples in $r \bowtie s$ is no greater than the number of tuples in s. The case where $R \cap S$ is a key for S is symmetric to the case just described. If $R \cap S$ forms a foreign key of S, referencing R, the number of tuples in $r \bowtie s$ is exactly the same as the number of tuples in s.
- The most difficult case is when $R \cap S$ is a key for neither R nor S. In this case, we assume, as we did for selections, that each value appears with equal probability. Consider a tuple t of r, and assume $R \cap S = \{A\}$. We estimate that tuple t produces

$$\frac{n_s}{V(A, s)}$$

tuples in $r \bowtie s$, since this number is the average number of tuples in s with a given value for the attributes A. Considering all the tuples in r, we estimate that there are

$$\frac{n_r * n_s}{V(A, s)}$$

tuples in $r \bowtie s$. Observe that, if we reverse the roles of r and s in the preceding estimate, we obtain an estimate of

$$\frac{n_r * n_s}{V(A, r)}$$

tuples in $r \bowtie s$. These two estimates differ if $V(A, r) \neq V(A, s)$. If this situation occurs, there are likely to be dangling tuples that do not participate in the join. Thus, the lower of the two estimates is probably the more accurate one.

The preceding estimate of join size may be too high if the $V(A, r)$ values for attribute A in r have few values in common with the $V(A, s)$ values for attribute A in s. However, this situation is unlikely to happen in the real world, since dangling tuples either do not exist or constitute only a small fraction of the tuples, in most real-world relations.

More important, the preceding estimate depends on the assumption that each value appears with equal probability. More sophisticated techniques for size estimation have to be used if this assumption does not hold. For example, if we have histograms on the join attributes of both relations, and both histograms have the same ranges, then we can use the above estimation technique within each range,

using the number of rows with values in the range instead of n_r or n_s, and the number of distinct values in that range, instead of $V(A, r)$ or $V(A, s)$. We then add up the size estimates obtained for each range to get the overall size estimate. We leave the case where both relations have histograms on the join attribute, but the histograms have different ranges, as an exercise for you.

We can estimate the size of a theta join $r \bowtie_\theta s$ by rewriting the join as $\sigma_\theta(r \times s)$ and using the size estimates for Cartesian products along with the size estimates for selections, which we saw in Section 16.3.2.

To illustrate all these ways of estimating join sizes, consider the expression:

$$student \bowtie takes$$

Assume the following catalog information about the two relations:

- $n_{student} = 5000$.
- $n_{takes} = 10000$.
- $V(ID, takes) = 2500$, which implies that only half the students have taken any course (this is unrealistic, but we use it to show that our size estimates are correct even in this case), and on average, each student who has taken a course has taken four courses.

Note that since *ID* is a primary key of *student*, $V(ID, student) = n_{student} = 5000$.

The attribute *ID* in *takes* is a foreign key on *student*, and null values do not occur in *takes.ID*, since *ID* is part of the primary key of *takes*; thus, the size of $student \bowtie takes$ is exactly n_{takes}, which is 10000.

We now compute the size estimates for $student \bowtie takes$ without using information about foreign keys. Since $V(ID, takes) = 2500$ and $V(ID, student) = 5000$, the two estimates we get are $5000 * 10000/2500 = 20000$ and $5000 * 10000/5000 = 10000$, and we choose the lower one. In this case, the lower of these estimates is the same as that which we computed earlier from information about foreign keys.

16.3.4 Size Estimation for Other Operations

Next we outline how to estimate the sizes of the results of other relational-algebra operations.

- **Projection:** The estimated size (number of records or number of tuples) of a projection of the form $\Pi_A(r)$ is $V(A, r)$, since projection eliminates duplicates.
- **Aggregation:** The size of ${}_G\gamma_A(r)$ is simply $V(G, r)$, since there is one tuple in ${}_G\gamma_A(r)$ for each distinct value of G.
- **Set operations:** If the two inputs to a set operation are selections on the same relation, we can rewrite the set operation as disjunctions, conjunctions, or negations. For example, $\sigma_{\theta_1}(r) \cup \sigma_{\theta_2}(r)$ can be rewritten as $\sigma_{\theta_1 \vee \theta_2}(r)$. Similarly, we can rewrite

intersections as conjunctions, and we can rewrite set difference by using negation, so long as the two relations participating in the set operations are selections on the same relation. We can then use the estimates for selections involving conjunctions, disjunctions, and negation in Section 16.3.2.

If the inputs are not selections on the same relation, we estimate the sizes this way: The estimated size of $r \cup s$ is the sum of the sizes of r and s. The estimated size of $r \cap s$ is the minimum of the sizes of r and s. The estimated size of $r - s$ is the same size as r. All three estimates may be inaccurate, but provide upper bounds on the sizes.

- **Outer join:** The estimated size of $r \rtimes s$ is the size of $r \bowtie s$ plus the size of r; that of $r \ltimes s$ is symmetric, while that of $r \rtimes\ltimes s$ is the size of $r \bowtie s$ plus the sizes of r and s. All three estimates may be inaccurate, but provide upper bounds on the sizes.

16.3.5 Estimation of Number of Distinct Values

The size estimates discussed earlier depend on statistics such as histograms, or at a minimum, the number of distinct values for an attribute. While these statistics can be precomputed and stored for relations in the database, we need to compute them for intermediate results. Note that estimation of the number of sizes and the number of distinct values of attributes in an intermediate result E_i helps us estimate the sizes and number of distinct values of attributes in the next level intermediate results that use E_i.

For selections, the number of distinct values of an attribute (or set of attributes) A in the result of a selection, $V(A, \sigma_\theta(r))$, can be estimated in these ways:

- If the selection condition θ forces A to take on a specified value (e.g., $A = 3$), $V(A, \sigma_\theta(r)) = 1$.
- If θ forces A to take on one of a specified set of values (e.g., $(A = 1 \vee A = 3 \vee A = 4)$), then $V(A, \sigma_\theta(r))$ is set to the number of specified values.
- If the selection condition θ is of the form A *op* v, where *op* is a comparison operator, $V(A, \sigma_\theta(r))$ is estimated to be $V(A, r) * s$, where s is the selectivity of the selection.
- In all other cases of selections, we assume that the distribution of A values is independent of the distribution of the values on which selection conditions are specified, and we use an approximate estimate of $\min(V(A, r), n_{\sigma_\theta(r)})$. A more accurate estimate can be derived for this case using probability theory, but the preceding approximation works fairly well.

For joins, the number of distinct values of an attribute (or set of attributes) A in the result of a join, $V(A, r \bowtie s)$, can be estimated in these ways:

- If all attributes in A are from r, $V(A, r \bowtie s)$ is estimated as $\min(V(A, r), n_{r \bowtie s})$, and similarly if all attributes in A are from s, $V(A, r \bowtie s)$ is estimated to be $\min(V(A, s), n_{r \bowtie s})$.
- If A contains attributes $A1$ from r and $A2$ from s, then $V(A, r \bowtie s)$ is estimated as:

$$\min(V(A1, r) * V(A2 - A1, s), V(A1 - A2, r) * V(A2, s), n_{r \bowtie s})$$

  Note that some attributes may be in $A1$ as well as in $A2$, and $A1 - A2$ and $A2 - A1$ denote, respectively, attributes in A that are only from r and attributes in A that are only from s. Again, more accurate estimates can be derived by using probability theory, but the above approximations work fairly well.

The estimates of distinct values are straightforward for projections: They are the same in $\Pi_A(r)$ as in r. The same holds for grouping attributes of aggregation. For results of **sum**, **count**, and **average**, we can assume, for simplicity, that all aggregate values are distinct. For **min**(A) and **max**(A), the number of distinct values can be estimated as $\min(V(A, r), V(G, r))$, where G denotes the grouping attributes. We omit details of estimating distinct values for other operations.

16.4 Choice of Evaluation Plans

Generation of expressions is only part of the query-optimization process, since each operation in the expression can be implemented with different algorithms. An evaluation plan defines exactly what algorithm should be used for each operation, and how the execution of the operations should be coordinated.

Given an evaluation plan, we can estimate its cost using statistics estimated by the techniques in Section 16.3 coupled with cost estimates for various algorithms and evaluation methods described in Chapter 15.

A **cost-based optimizer** explores the space of all query-evaluation plans that are equivalent to the given query, and chooses the one with the least estimated cost. We have seen how equivalence rules can be used to generate equivalent plans. However, cost-based optimization with arbitrary equivalence rules is fairly complicated. We first cover a simpler version of cost-based optimization, which involves only join-order and join algorithm selection, in Section 16.4.1. Then, in Section 16.4.2, we briefly sketch how a general-purpose optimizer based on equivalence rules can be built, without going into details.

Exploring the space of all possible plans may be too expensive for complex queries. Most optimizers include heuristics to reduce the cost of query optimization, at the potential risk of not finding the optimal plan. We study some such heuristics in Section 16.4.3.

16.4.1 Cost-Based Join-Order Selection

The most common type of query in SQL consists of a join of a few relations, with join predicates and selections specified in the **where** clause. In this section we consider the problem of choosing the optimal join order for such a query.

For a complex join query, the number of different query plans that are equivalent to the query can be large. As an illustration, consider the expression:

$$r_1 \bowtie r_2 \bowtie \cdots \bowtie r_n$$

where the joins are expressed without any ordering. With $n = 3$, there are 12 different join orderings:

$$\begin{array}{llll} r_1 \bowtie (r_2 \bowtie r_3) & r_1 \bowtie (r_3 \bowtie r_2) & (r_2 \bowtie r_3) \bowtie r_1 & (r_3 \bowtie r_2) \bowtie r_1 \\ r_2 \bowtie (r_1 \bowtie r_3) & r_2 \bowtie (r_3 \bowtie r_1) & (r_1 \bowtie r_3) \bowtie r_2 & (r_3 \bowtie r_1) \bowtie r_2 \\ r_3 \bowtie (r_1 \bowtie r_2) & r_3 \bowtie (r_2 \bowtie r_1) & (r_1 \bowtie r_2) \bowtie r_3 & (r_2 \bowtie r_1) \bowtie r_3 \end{array}$$

In general, with n relations, there are $(2(n-1))!/(n-1)!$ different join orders. (We leave the computation of this expression for you to do in Exercise 16.12.) For joins involving small numbers of relations, this number is acceptable; for example, with $n = 5$, the number is 1680. However, as n increases, this number rises quickly. With $n = 7$, the number is 665,280; with $n = 10$, the number is greater than 17.6 billion!

Luckily, it is not necessary to generate all the expressions equivalent to a given expression. For example, suppose we want to find the best join order of the form:

$$(r_1 \bowtie r_2 \bowtie r_3) \bowtie r_4 \bowtie r_5$$

which represents all join orders where r_1, r_2, and r_3 are joined first (in some order), and the result is joined (in some order) with r_4 and r_5. There are 12 different join orders for computing $r_1 \bowtie r_2 \bowtie r_3$, and 12 orders for computing the join of this result with r_4 and r_5. Thus, there appear to be 144 join orders to examine. However, once we have found the best join order for the subset of relations $\{r_1, r_2, r_3\}$, we can use that order for further joins with r_4 and r_5, and we can ignore all costlier join orders of $r_1 \bowtie r_2 \bowtie r_3$. Thus, instead of 144 choices to examine, we need to examine only 12 + 12 choices.

Using this idea, we can develop a *dynamic-programming* algorithm for finding optimal join orders. Dynamic-programming algorithms store results of computations and reuse them, a procedure that can reduce execution time greatly.

We now consider how to find the optimal join order for a set of n relations $S = \{r_1, r_2, \ldots, r_n\}$, where each relation may have selection conditions, and a set of join conditions between the relations r_i is provided. We assume that relations have unique names.

A recursive procedure implementing the dynamic-programming algorithm appears in Figure 16.7 and is invoked as FindBestPlan(S), where S is the set of relations above. The procedure applies selections on individual relations at the earliest possible point,

```
procedure FindBestPlan(S)
    if (bestplan[S].cost ≠ ∞) /* bestplan[S] already computed */
        return bestplan[S]
    if (S contains only 1 relation)
        set bestplan[S].plan and bestplan[S].cost based on the best way of
            accessing S using selection conditions (if any) on S.
    else for each non-empty subset S1 of S such that S1 ≠ S
        P1 = FindBestPlan(S1)
        P2 = FindBestPlan(S − S1)
        for each algorithm A for joining the results of P1 and P2
            // For indexed-nested loops join, the outer relation could be P1 or P2.
            // Similarly for hash-join, the build relation could be P1 or P2.
            // We assume the alternatives are considered as separate algorithms.
            // We assume cost of A does not include cost of reading the inputs.
            if algorithm A is indexed nested loops
                Let P_o and P_i denote the outer and inner inputs of A
                if P_i has a single relation r_i, and r_i has an index on the join
                        attributes
                    plan = "execute P_o.plan; join results of P_o and r_i using A",
                            with any selection condition on P_i performed as
                            part of the join condition
                    cost = P_o.cost + cost of A
                else /* Cannot use indexed nested loops join */
                    cost = ∞
            else
                plan = "execute P1.plan; execute P2.plan;
                            join results of P1 and P2 using A"
                cost = P1.cost + P2.cost + cost of A
            if cost < bestplan[S].cost
                bestplan[S].cost = cost
                bestplan[S].plan = plan
    return bestplan[S]
```

Figure 16.7 Dynamic-programming algorithm for join-order optimization.

that is, when the relations are accessed. It is easiest to understand the procedure assuming that all joins are natural joins, although the procedure works unchanged with any join condition. With arbitrary join conditions, the join of two subexpressions is understood to include all join conditions that relate attributes from the two subexpressions.

The procedure stores the evaluation plans it computes in an associative array *bestplan*, which is indexed by sets of relations. Each element of the associative array contains two components: the cost of the best plan of S, and the plan itself. The value of $bestplan[S].cost$ is assumed to be initialized to ∞ if $bestplan[S]$ has not yet been computed.

The procedure first checks if the best plan for computing the join of the given set of relations S has been computed already (and stored in the associative array *bestplan*); if so, it returns the already computed plan.

If S contains only one relation, the best way of accessing S (taking selections on S, if any, into account) is recorded in *bestplan*. This may involve using an index to identify tuples, and then fetching the tuples (often referred to as an *index scan*), or scanning the entire relation (often referred to as a *relation scan*).[2] If there is any selection condition on S, other than those ensured by an index scan, a selection operation is added to the plan to ensure all selections on S are satisfied.

Otherwise, if S contains more than one relation, the procedure tries every way of dividing S into two disjoint subsets. For each division, the procedure recursively finds the best plans for each of the two subsets. It then considers all possible algorithms for joining the results of the two subsets. Note that since indexed nested loops join can potentially use either input $P1$ or $P2$ as the inner input, we consider the two alternatives as two different algorithms. The choice of build versus probe input also leads us to consider the two choices for hash join as two different algorithms.

The cost of each alternative is considered, and the least cost option chosen. The join cost considered should not include the cost of reading the inputs, since we assume that the input is pipelined from the preceding operators, which could be a relation/index scan, or a preceding join. Recall that some operators, such as hash join, can be treated as having suboperators with a blocking (materialized) edge between them, but with the input and output edges of the join being pipelined. The join cost formulae that we saw in Chapter 15 can be used with appropriate modifications to ignore the cost of reading the input relations. Note that indexed nested loops join is treated differently from other join techniques: the plan as well as the cost are different in this case, since we do not perform a relation/index scan of the inner input, and the index lookup cost is included in the cost of indexed nested loops join.

The procedure picks the cheapest plan from among all the alternatives for dividing S into two sets and the algorithms for joining the results of the two sets. The cheapest plan and its cost are stored in the array *bestplan* and returned by the procedure. The time complexity of the procedure can be shown to be $O(3^n)$ (see Practice Exercise 16.13).

The order in which tuples are generated by the join of a set of relations is important for finding the best overall join order, since it can affect the cost of further joins. For

[2] If an index contains all the attributes of a relation that are used in a query, it is possible to perform an *index-only scan*, which retrieves the required attribute values from the index, without fetching actual tuples.

instance, if merge join is used, a potentially expensive sort operation is required on the input, unless the input is already sorted on the join attribute.

A particular sort order of the tuples is said to be an **interesting sort order** if it could be useful for a later operation. For instance, generating the result of $r_1 \bowtie r_2 \bowtie r_3$ sorted on the attributes common with r_4 or r_5 may be useful, but generating it sorted on the attributes common to only r_1 and r_2 is not useful. Using merge join for computing $r_1 \bowtie r_2 \bowtie r_3$ may be costlier than using some other join technique, but it may provide an output sorted in an interesting sort order.

Hence, it is not sufficient to find the best join order for each subset of the set of n given relations. Instead, we have to find the best join order for each subset, for each interesting sort order of the join result for that subset. The *bestplan* array can now be indexed by $[S, o]$, where S is a set of relations, and o is an interesting sort order. The FindBestPlan function can then be modified to take interesting sort orders into consideration; we leave details as an exercise for you (see Practice Exercise 16.11).

The number of subsets of n relations is 2^n. The number of interesting sort orders is generally not large. Thus, about 2^n join expressions need to be stored. The dynamic-programming algorithm for finding the best join order can be extended to handle sort orders. Specifically, when considering sort-merge join, the cost of sorting has to be added if an input (which may be a relation, or the result of a join operation) is not sorted on the join attribute, but is not added if it is sorted.

The cost of the extended algorithm depends on the number of interesting orders for each subset of relations; since this number has been found to be small in practice, the cost remains at $O(3^n)$. With $n = 10$, this number is around 59,000, which is much better than the 17.6 billion different join orders. More important, the storage required is much less than before, since we need to store only one join order for each interesting sort order of each of 1024 subsets of $r_1, \ldots, r_{10}$. Although both numbers still increase rapidly with n, commonly occurring joins usually have less than 10 relations and can be handled easily.

The code shown in Figure 16.7 actually considers each possible way of dividing S into two disjoint subsets twice, since each of the two subsets can play the role of $S1$. Considering the division twice does not affect correctness, but wastes time. The code can be optimized as follows: find the alphabetically smallest relation r_i in $S1$, and the alphabetically smallest relation r_j in $S - S1$, and execute the loop only if $r_i < r_j$. Doing so ensures that each division is considered only once.

Further, the code also considers all possible join orders, including those that contain Cartesian products; for example, if two relations r_1 and r_3 do not have any join condition linking the two relations, the code will still consider $S = \{r_1, r_3\}$, which will result in a Cartesian product. It is possible to take join conditions into account, and modify the code to only generate divisions that do not result in Cartesian products. This optimization can save a great deal of time for many queries. See the Further Reading section at the end of the chapter for references providing more details on Cartesian-product-free join order enumeration.

16.4.2 Cost-Based Optimization with Equivalence Rules

The join-order optimization technique we just saw handles the most common class of queries, which perform an inner join of a set of relations. However, many queries use other features, such as aggregation, outer join, and nested queries, which are not addressed by join-order selection, but can be handled by using equivalence rules.

In this section we outline how to create a general-purpose cost-based optimizer based on equivalence rules. Equivalence rules can help explore alternatives with a wide variety of operations, such as outer joins, aggregations, and set operations, as we have seen earlier. Equivalence rules can be added if required for further operations, such as operators that return the top-K results in sorted order.

In Section 16.2.4, we saw how an optimizer could systematically generate all expressions equivalent to the given query. The procedure for generating equivalent expressions can be modified to generate all possible evaluation plans as follows: A new class of equivalence rules, called **physical equivalence rules**, is added that allows a logical operation, such as a join, to be transformed to a physical operation, such as a hash join, or a nested-loops join. By adding such rules to the original set of equivalence rules, the procedure can generate all possible evaluation plans. The cost estimation techniques we have seen earlier can then be used to choose the optimal (i.e., the least-cost) plan.

However, the procedure shown in Section 16.2.4 is very expensive, even if we do not consider generation of evaluation plans. To make the approach work efficiently requires the following:

1. A space-efficient representation of expressions that avoids making multiple copies of the same subexpressions when equivalence rules are applied.
2. E fficient techniques for detecting duplicate derivations of the same expression.
3. A form of dynamic programming based on **memoization**, which stores the optimal query evaluation plan for a subexpression when it is optimized for the first time; subsequent requests to optimize the same subexpression are handled by returning the already memorized plan.
4. Techniques that avoid generating all possible equivalent plans by keeping track of the cheapest plan generated for any subexpression up to any point of time, and pruning away any plan that is more expensive than the cheapest plan found so far for that subexpression.

The details are more complex than we wish to deal with here. This approach was pioneered by the Volcano research project, and the query optimizer of SQL Server is based on this approach. See the bibliographical notes for references containing further information.

16.4.3 Heuristics in Optimization

A drawback of cost-based optimization is the cost of optimization itself. Although the cost of query optimization can be reduced by clever algorithms, the number of different

evaluation plans for a query can be very large, and finding the optimal plan from this set requires a lot of computational effort. Hence, optimizers use **heuristics** to reduce the cost of optimization.

An example of a heuristic rule is the following rule for transforming relational-algebra queries:

- Perform selection operations as early as possible.

A heuristic optimizer would use this rule without finding out whether the cost is reduced by this transformation. In the first transformation example in Section 16.2, the selection operation was pushed into a join.

We say that the preceding rule is a heuristic because it usually, but not always, helps to reduce the cost. For an example of where it can result in an increase in cost, consider an expression $\sigma_\theta(r \bowtie s)$, where the condition θ refers to only attributes in s. The selection can certainly be performed before the join. However, if r is extremely small compared to s, and if there is an index on the join attributes of s, but no index on the attributes used by θ, then it is probably a bad idea to perform the selection early. Performing the selection early—that is, directly on s—would require doing a scan of all tuples in s. It is probably cheaper, in this case, to compute the join by using the index and then to reject tuples that fail the selection. (This case is specifically handled by the dynamic programming algorithm for join order optimization.)

The projection operation, like the selection operation, reduces the size of relations. Thus, whenever we need to generate a temporary relation, it is advantageous to apply immediately any projections that are possible. This advantage suggests a companion to the "perform selections early" heuristic:

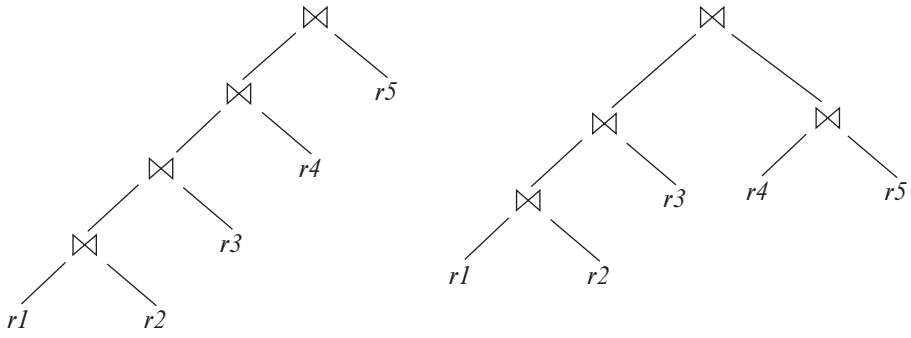
- Perform projections early.

It is usually better to perform selections earlier than projections, since selections have the potential to reduce the sizes of relations greatly, and selections enable the use of indices to access tuples. An example similar to the one used for the selection heuristic should convince you that this heuristic does not always reduce the cost.

Optimizers based on join-order enumeration typically use heuristic transformations to handle constructs other than joins, and applying the cost-based join-order selection algorithm to subexpressions involving only joins and selections. Details of such heuristics are for the most part specific to individual optimizers, and we do not cover them.

Most practical query optimizers have further heuristics to reduce the cost of optimization. For example, many query optimizers, such as the System R optimizer,[3] do not consider all join orders, but rather restrict the search to particular kinds of join or-

[3]System R was one of the first implementations of SQL, and its optimizer pioneered the idea of cost-based join-order optimization.



(a) Left-deep join tree (b) Non-left-deep join tree

Figure 16.8 Left-deep join trees.

ders. The System R optimizer considers only those join orders where the right operand of each join is one of the initial relations $r_1, \ldots, r_n$. Such join orders are called **left-deep join orders**. Left-deep join orders are particularly convenient for pipelined evaluation, since the right operand is a stored relation, and thus only one input to each join is pipelined.

Figure 16.8 illustrates the difference between left-deep join trees and non-left-deep join trees. The time it takes to consider all left-deep join orders is $O(n!)$, which is much less than the time to consider all join orders. With the use of dynamic-programming optimizations, the System R optimizer can find the best join order in time $O(n2^n)$. Contrast this cost with the $O(3^n)$ time required to find the best overall join order. The System R optimizer uses heuristics to push selections and projections down the query tree.

A heuristic approach to reduce the cost of join-order selection, which was originally used in some versions of Oracle, works roughly this way: For an n-way join, it considers n evaluation plans. Each plan uses a left-deep join order, starting with a different one of the n relations. The heuristic constructs the join order for each of the n evaluation plans by repeatedly selecting the "best" relation to join next, on the basis of a ranking of the available access paths. Either nested-loop or sort-merge join is chosen for each of the joins, depending on the available access paths. Finally, the heuristic chooses one of the n evaluation plans in a heuristic manner, on the basis of minimizing the number of nested-loop joins that do not have an index available on the inner relation and on the number of sort-merge joins.

Query-optimization approaches that apply heuristic plan choices for some parts of the query, with cost-based choice based on generation of alternative access plans on other parts of the query, have been adopted in several systems. The approach used in System R and in its successor, the Starburst project, is a hierarchical procedure based on the nested-block concept of SQL. The cost-based optimization techniques described here are used for each block of the query separately. The optimizers in several

database products, such as IBM DB2 and Oracle, are based on the above approach, with extensions to handle other operations such as aggregation. For compound SQL queries (using the ∪, ∩, or − operation), the optimizer processes each component separately and combines the evaluation plans to form the overall evaluation plan.

Most optimizers allow a cost budget to be specified for query optimization. The search for the optimal plan is terminated when the **optimization cost budget** is exceeded, and the best plan found up to that point is returned. The budget itself may be set dynamically; for example, if a cheap plan is found for a query, the budget may be reduced, on the premise that there is no point spending a lot of time optimizing the query if the best plan found so far is already quite cheap. On the other hand, if the best plan found so far is expensive, it makes sense to invest more time in optimization, which could result in a significant reduction in execution time. To best exploit this idea, optimizers usually first apply cheap heuristics to find a plan and then start full cost-based optimization with a budget based on the heuristically chosen plan.

Many applications execute the same query repeatedly, but with different values for the constants. For example, a university application may repeatedly execute a query to find the courses for which a student has registered, but each time for a different student with a different value for the student ID. As a heuristic, many optimizers optimize a query once, with whatever values were provided for the constants when the query was first submitted, and cache the query plan. Whenever the query is executed again, perhaps with new values for constants, the cached query plan is reused (using new values for the constants). The optimal plan for the new constants may differ from the optimal plan for the initial values, but as a heuristic the cached plan is reused.[4] Caching and reuse of query plans is referred to as **plan caching**.

Even with the use of heuristics, cost-based query optimization imposes a substantial overhead on query processing. However, the added cost of cost-based query optimization is usually more than offset by the saving at query-execution time, which is dominated by slow disk accesses. The difference in execution time between a good plan and a bad one may be huge, making query optimization essential. The achieved saving is magnified in those applications that run on a regular basis, where a query can be optimized once, and the selected query plan can be used each time the query is executed. Therefore, most commercial systems include relatively sophisticated optimizers. The bibliographical notes give references to descriptions of the query optimizers of actual database systems.

16.4.4 Optimizing Nested Subqueries

SQL conceptually treats nested subqueries in the **where** clause as functions that take parameters and return either a single value or a set of values (possibly an empty set). The parameters are the variables from an outer-level query that are used in the nested

[4]For the student registration query, the plan would almost certainly be the same for any student ID. But a query that took a range of student IDs, and returned registration information for all student IDs in that range, would probably have a different optimal plan if the range were very small than if the range were large.

subquery (these variables are called **correlation variables**). For instance, suppose we have the following query, to find the names of all instructors who taught a course in 2019:

```
select name
from instructor
where exists (select *
              from teaches
              where instructor.ID = teaches.ID
                    and teaches.year = 2019);
```

Conceptually, the subquery can be viewed as a function that takes a parameter (here, *instructor.ID*) and returns the set of all courses taught in 2019 by instructors (with the same *ID*).

SQL evaluates the overall query (conceptually) by computing the Cartesian product of the relations in the outer **from** clause and then testing the predicates in the **where** clause for each tuple in the product. In the preceding example, the predicate tests if the result of the subquery evaluation is empty. In practice, the predicates in the **where** clause that can be used as join predicates, or as selection predicates are evaluated as part of the selections on relations or to perform joins that avoid Cartesian products. Predicates involving nested subqueries in the **where** clause are evaluated subsequently, since they are usually expensive, by invoking the subquery as a function.

The technique of evaluating a nested subquery by invoking it as a function is called **correlated evaluation**. Correlated evaluation is not very efficient, since the subquery is separately evaluated for each tuple in the outer level query. A large number of random disk I/O operations may result.

SQL optimizers therefore attempt to transform nested subqueries into joins, where possible. Efficient join algorithms help avoid expensive random I/O. Where the transformation is not possible, the optimizer keeps the subqueries as separate expressions, optimizes them separately, and then evaluates them by correlated evaluation.

As an attempt at transforming a nested subquery into a join, the query in the preceding example can be rewritten in relational algebra as a join:

$$\Pi_{name}(instructor \bowtie_{instructor.ID=teaches.ID \wedge teaches.year=2019} teaches)$$

Unfortunately, the above query is not quite correct, since the multiset versions of the relational algebra operators are used in SQL implementations, and as a result an instructor who teaches multiple sections in 2019 will appear multiple times in the result of the relational algebra query, although that instructor would appear only once in the SQL query result. Using the set version of the relational algebra operators will not help either, since if there are two instructors with the same name who teach in 2019, the name would appear only once with the set version of relational algebra, but would appear twice in the SQL query result. (We note that the set version of relational algebra

would give the correct result if the query output contained the primary key of *instructor*, namely *ID*.)

To properly reflect SQL semantics, the number of duplicates of a tuple in the result should not change because of the rewriting. The semijoin operator of the relational algebra provides a solution to this problem. The multiset version of the **semijoin** operator $r \ltimes_\theta s$ is defined as follows: if a tuple r_i appears n times in r, it appears n times in the result of $r \ltimes_\theta$ if there is at least one tuple s_j such that r_i and s_j together satisfy predicate θ; otherwise r_i does not appear in the result. The set version of the semijoin operator $r \ltimes_\theta s$ can be defined as $\Pi_R(r \bowtie_\theta s)$, where R is the set of attributes in the schema of r. The multiset version of the semijoin operator outputs the same tuples, but the number of duplicates of each tuple r_i in the semijoin result is the same as the number of duplicates of r_i in r.

The preceding SQL query can be translated into the following equivalent relational algebra using the multiset semijoin operator:

$$\Pi_{name}(instructor \ltimes_{instructor.ID=teaches.ID \wedge teaches.year=2019} teaches)$$

The above query in the multiset relational algebra gives the same result as the SQL query, including the counts of duplicates. The query can equivalently be written as:

$$\Pi_{name}(instructor \ltimes_{instructor.ID=teaches.ID} (\sigma_{teaches.year=2019}(teaches)))$$

The following SQL query using the **in** clause is equivalent to the preceding SQL query using the **exists** clause, and can be translated to the same relational algebra expression using semijoin.

```
select name
from instructor
where instructor.ID in (select teaches.ID
                        from teaches
                        where teaches.year = 2019);
```

The anti-semijoin is useful with **not exists** queries. The multiset **anti-semijoin** operator $r \overline{\ltimes}_\theta s$ is defined as follows: if a tuple r_i appears n times in r, it appears n times in the result of $r \ltimes_\theta s$ if there does not exist any tuple s_j in s such that r_i and s_j satisfy predicate θ; otherwise r_i does not appear in the result. The anti-semijoin operator is also known as the **anti-join** operator.

Consider the SQL query:

```
select name
from instructor
where not exists (select *
                  from teaches
                  where instructor.ID = teaches.ID
                        and teaches.year = 2019);
```

The preceding query can be translated into the following relational algebra using the anti-semijoin operator:

$$\Pi_{name}(instructor \overline{\ltimes}_{instructor.ID=teaches.ID}(\sigma_{teaches.year=2019}(teaches)))$$

In general, a query of the form:

```
select A
from r1, r2, ..., rn
where P1 and exists (select *
                     from s1, s2, ..., sm
                     where P2^1 and P2^2);
```

where P_2^1 are predicates that only reference the relations s_i in the subquery, and P_2^2 predicates that also reference the relations r_i from the outer query, can be translated to:

$$\Pi_A((\sigma_{P_1}(r_1 \times r_2 \times \ldots \times r_n)) \ltimes_{P_2^2} \sigma_{P_2^1}(s_1 \times s_2 \times \ldots \times s_m))$$

If **not exists** were used instead of **exists**, the semijoin should be replaced by anti-semijoin in the relational algebra query. If an **in** clause is used instead of **exists**, the relational algebra query can be appropriately modified by adding a corresponding predicate in the semijoin predicate, as our earlier example illustrated.

The process of replacing a nested query by a query with a join, semijoin, or anti-semijoin is called **decorrelation**. The semijoin and anti-semijoin operators can be efficiently implemented using modifications of the join algorithms, as explored in Practice Exercise 15.10.

Consider the following query with aggregation in a scalar subquery, that finds instructors who have taught more than one course section in 2019.

```
select name
from instructor
where 1 < (select count(*)
           from teaches
           where instructor.ID = teaches.ID
                 and teaches.year = 2019);
```

The above query can be rewritten using a semijoin as follows:

$$\Pi_{name}(instructor \ltimes_{(instructor.ID=TID)\wedge(1<cnt)}({}_{ID\ \mathbf{as}\ TID}\gamma_{\mathbf{count}(*)\ \mathbf{as}\ cnt}(\sigma_{year=2019}(teaches))))$$

Observe that the subquery has a predicate *instructor.ID*= *teaches.ID*, and aggregation without a group by clause. The decorrelated query has the predicate moved into the semijoin condition, and the aggregation is now grouped by *ID*. The predicate 1 < (subquery) has turned into a semijoin predicate. Intuitively, the subquery performs a sep-

arate count for each *instructor.ID*; grouping by *ID* ensures that counts are computed separately for each *ID*.

Decorrelation is clearly more complicated when the nested subquery uses aggregation, or when the nested subquery is used as a scalar subquery. In fact, decorrelation is not possible for certain cases of subqueries. For example, a subquery that is used as a scalar subquery is expected to return only one result; if it returns more than one result, a runtime exception can occur, which is not possible with a decorrelated query. Further, whether to decorrelate or not should ideally be done in a cost-based manner, depending on whether decorrelation reduces the cost or not. Some query optimizers represent nested subqueries using extended relational-algebra constructs, and express transformations of nested subqueries to semijoin, anti-semijoin, and so forth, as equivalence rules. We do not attempt to give algorithms for the general case, and instead refer you to relevant items in the online bibliographical notes.

Optimization of complex nested subqueries is a complicated task, as you can infer from the preceding discussion, and many optimizers do only a limited amount of decorrelation. It better to avoid using complex nested subqueries, where possible, since we cannot be sure that the query optimizer will succeed in converting them to a form that can be evaluated efficiently.

16.5 Materialized Views

When a view is defined, normally the database stores only the query defining the view. In contrast, a **materialized view** is a view whose contents are computed and stored. Materialized views constitute redundant data, in that their contents can be inferred from the view definition and the rest of the database contents. However, it is much cheaper in many cases to read the contents of a materialized view than to compute the contents of the view by executing the query defining the view.

Materialized views are important for improving performance in some applications. Consider this view, which gives the total salary in each department:

```
create view department_total_salary(dept_name, total_salary) as
select dept_name, sum (salary)
from instructor
group by dept_name;
```

Suppose the total salary amount at a department is required frequently. Computing the view requires reading every *instructor* tuple pertaining to a department and summing up the salary amounts, which can be time-consuming. In contrast, if the view definition of the total salary amount were materialized, the total salary amount could be found by looking up a single tuple in the materialized view.[5]

[5]The difference may not be all that large for a medium-sized university, but in other settings the difference can be very large. For example, if the materialized view computed total sales of each product, from a sales relation with tens

16.5.1 View Maintenance

A problem with materialized views is that they must be kept up-to-date when the data used in the view definition changes. For instance, if the *salary* value of an instructor is updated, the materialized view will become inconsistent with the underlying data, and it must be updated. The task of keeping a materialized view up-to-date with the underlying data is known as **view maintenance**.

Views can be maintained by manually written code: That is, every piece of code that updates the *salary* value can be modified to also update the total salary amount for the corresponding department. However, this approach is error prone, since it is easy to miss some places where the *salary* is updated, and the materialized view will then no longer match the underlying data.

Another option for maintaining materialized views is to define triggers on insert, delete, and update of each relation in the view definition. The triggers must modify the contents of the materialized view, to take into account the change that caused the trigger to fire. A simplistic way of doing so is to completely recompute the materialized view on every update.

A better option is to modify only the affected parts of the materialized view, which is known as **incremental view maintenance**. We describe how to perform incremental view maintenance in Section 16.5.2.

Modern database systems provide more direct support for incremental view maintenance. Database-system programmers no longer need to define triggers for view maintenance. Instead, once a view is declared to be materialized, the database system computes the contents of the view and incrementally updates the contents when the underlying data change.

Most database systems perform **immediate view maintenance**; that is, incremental view maintenance is performed as soon as an update occurs, as part of the updating transaction. Some database systems also support **deferred view maintenance**, where view maintenance is deferred to a later time; for example, updates may be collected throughout a day, and materialized views may be updated at night. This approach reduces the overhead on update transactions. However, materialized views with deferred view maintenance may not be consistent with the underlying relations on which they are defined.

16.5.2 Incremental View Maintenance

To understand how to maintain materialized views incrementally, we start off by considering individual operations, and then we see how to handle a complete expression.

The changes to a relation that can cause a materialized view to become out-of-date are inserts, deletes, and updates. To simplify our description, we replace updates to a tuple by deletion of the tuple followed by insertion of the updated tuple. Thus, we need

---

of millions of tuples, the difference between computing the aggregate from the underlying data and looking up the materialized view can be many orders of magnitude.

to consider only inserts and deletes. The changes (inserts and deletes) to a relation or expression are referred to as its **differential**.

16.5.2.1 Join Operation

Consider the materialized view $v = r \bowtie s$. Suppose we modify r by inserting a set of tuples denoted by i_r. If the old value of r is denoted by r^{old}, and the new value of r by r^{new}, $r^{new} = r^{old} \cup i_r$. Now, the old value of the view, v^{old}, is given by $r^{old} \bowtie s$, and the new value v^{new} is given by $r^{new} \bowtie s$. We can rewrite $r^{new} \bowtie s$ as $(r^{old} \cup i_r) \bowtie s$, which we can again rewrite as $(r^{old} \bowtie s) \cup (i_r \bowtie s)$. In other words:

$$v^{new} = v^{old} \cup (i_r \bowtie s)$$

Thus, to update the materialized view v, we simply need to add the tuples $i_r \bowtie s$ to the old contents of the materialized view. Inserts to s are handled in an exactly symmetric fashion.

Now suppose r is modified by deleting a set of tuples denoted by d_r. Using the same reasoning as above, we get:

$$v^{new} = v^{old} - (d_r \bowtie s)$$

Deletes on s are handled in an exactly symmetric fashion.

16.5.2.2 Selection and Projection Operations

Consider a view $v = \sigma_\theta(r)$. If we modify r by inserting a set of tuples i_r, the new value of v can be computed as:

$$v^{new} = v^{old} \cup \sigma_\theta(i_r)$$

Similarly, if r is modified by deleting a set of tuples d_r, the new value of v can be computed as:

$$v^{new} = v^{old} - \sigma_\theta(d_r)$$

Projection is a more difficult operation with which to deal. Consider a materialized view $v = \Pi_A(r)$. Suppose the relation r is on the schema $R = (A, B)$, and r contains two tuples $(a, 2)$ and $(a, 3)$. Then, $\Pi_A(r)$ has a single tuple (a). If we delete the tuple $(a, 2)$ from r, we cannot delete the tuple (a) from $\Pi_A(r)$: If we did so, the result would be an empty relation, whereas in reality $\Pi_A(r)$ still has a single tuple (a). The reason is that the same tuple (a) is derived in two ways, and deleting one tuple from r removes only one of the ways of deriving (a); the other is still present.

This reason also gives us the intuition for a solution: For each tuple in a projection such as $\Pi_A(r)$, we will keep a count of how many times it was derived.

When a set of tuples d_r is deleted from r, for each tuple t in d_r we do the following: Let $t.A$ denote the projection of t on the attribute A. We find $(t.A)$ in the materialized view and decrease the count stored with it by 1. If the count becomes 0, $(t.A)$ is deleted from the materialized view.

Handling insertions is relatively straightforward. When a set of tuples i_r is inserted into r, for each tuple t in i_r we do the following: If $(t.A)$ is already present in the materialized view, we increase the count stored with it by 1. If not, we add $(t.A)$ to the materialized view, with the count set to 1.

16.5.2.3 Aggregation Operations

Aggregation operations proceed somewhat like projections. The aggregate operations in SQL are **count, sum, avg, min,** and **max**:

- **count**: Consider a materialized view $v = {}_G\gamma_{count(B)}(r)$, which computes the count of the attribute B, after grouping r by attribute G.

  When a set of tuples i_r is inserted into r, for each tuple t in i_r we do the following: We look for the group $t.G$ in the materialized view. If it is not present, we add $(t.G, 1)$ to the materialized view. If the group $t.G$ is present, we add 1 to the count of the group.

  When a set of tuples d_r is deleted from r, for each tuple t in d_r we do the following: We look for the group $t.G$ in the materialized view and subtract 1 from the count for the group. If the count becomes 0, we delete the tuple for the group $t.G$ from the materialized view.

- **sum**: Consider a materialized view $v = {}_G\gamma_{sum(B)}(r)$.
  When a set of tuples i_r is inserted into r, for each tuple t in i_r we do the following: We look for the group $t.G$ in the materialized view. If it is not present, we add $(t.G, t.B)$ to the materialized view; in addition, we store a count of 1 associated with $(t.G, t.B)$, just as we did for projection. If the group $t.G$ is present, we add the value of $t.B$ to the aggregate value for the group and add 1 to the count of the group.

  When a set of tuples d_r is deleted from r, for each tuple t in d_r we do the following: We look for the group $t.G$ in the materialized view and subtract $t.B$ from the aggregate value for the group. We also subtract 1 from the count for the group, and if the count becomes 0, we delete the tuple for the group $t.G$ from the materialized view.

  Without keeping the extra count value, we would not be able to distinguish a case where the sum for a group is 0 from the case where the last tuple in a group is deleted.

- **avg**: Consider a materialized view $v = {}_G\gamma_{avg(B)}(r)$.
  Directly updating the average on an insert or delete is not possible, since it depends not only on the old average and the tuple being inserted/deleted, but also on the number of tuples in the group.

Instead, to handle the case of **avg**, we maintain the **sum** and **count** aggregate values as described earlier and compute the average as the sum divided by the count.

- **min, max**: Consider a materialized view $v = {}_G\gamma_{min(B)}(r)$. (The case of **max** is exactly equivalent.)

  Handling insertions on r is straightforward, similar to the case of **sum**. Maintaining the aggregate values **min** and **max** on deletions may be more expensive. For example, if the tuple t corresponding to the minimum value for a group is deleted from r, we have to look at the other tuples of r that are in the same group to find the new minimum value. It is a good idea to create an ordered index on (G, B) since it would help us to find the new minimum value for a group very efficiently.

16.5.2.4 Other Operations

The set operation *intersection* is maintained as follows: Given materialized view $v = r \cap s$, when a tuple is inserted in r we check if it is present in s, and if so we add it to v. If a tuple is deleted from r, we delete it from the intersection if it is present. The other set operations, *union* and *set difference*, are handled in a similar fashion; we leave details to you.

Outer joins are handled in much the same way as joins, but with some extra work. In the case of deletion from r we have to handle tuples in s that no longer match any tuple in r. In the case of insertion to r, we have to handle tuples in s that did not match any tuple in r. Again we leave details to you.

16.5.2.5 Handling Expressions

So far we have seen how to update incrementally the result of a single operation. To handle an entire expression, we can derive expressions for computing the incremental change to the result of each subexpression, starting from the smallest subexpressions.

For example, suppose we wish to incrementally update a materialized view $E_1 \bowtie E_2$ when a set of tuples i_r is inserted into relation r. Let us assume r is used in E_1 alone. Suppose the set of tuples to be inserted into E_1 is given by expression D_1. Then the expression $D_1 \bowtie E_2$ gives the set of tuples to be inserted into $E_1 \bowtie E_2$.

See the online bibliographical notes for further details on incremental view maintenance with expressions.

16.5.3 Query Optimization and Materialized Views

Query optimization can be performed by treating materialized views just like regular relations. However, materialized views offer further opportunities for optimization:

- Rewriting queries to use materialized views:

  Suppose a materialized view $v = r \bowtie s$ is available, and a user submits a query $r \bowtie s \bowtie t$. Rewriting the query as $v \bowtie t$ may provide a more efficient query plan

than optimizing the query as submitted. Thus, it is the job of the query optimizer to recognize when a materialized view can be used to speed up a query.

- Replacing a use of a materialized view with the view definition:
  Suppose a materialized view $v = r \bowtie s$ is available, but without any index on it, and a user submits a query $\sigma_{A=10}(v)$. Suppose also that s has an index on the common attribute B, and r has an index on attribute A. The best plan for this query may be to replace v with $r \bowtie s$, which can lead to the query plan $\sigma_{A=10}(r) \bowtie s$; the selection and join can be performed efficiently by using the indices on $r.A$ and $s.B$, respectively. In contrast, evaluating the selection directly on v may require a full scan of v, which may be more expensive.

The online bibliographical notes give pointers to research showing how to perform query optimization efficiently with materialized views.

16.5.4 Materialized View and Index Selection

Another related optimization problem is that of **materialized view selection**, namely, "What is the best set of views to materialize?" This decision must be made on the basis of the system **workload**, which is a sequence of queries and updates that reflects the typical load on the system. One simple criterion would be to select a set of materialized views that minimizes the overall execution time of the workload of queries and updates, including the time taken to maintain the materialized views. Database administrators usually modify this criterion to take into account the importance of different queries and updates: Fast response may be required for some queries and updates, but a slow response may be acceptable for others.

Indices are just like materialized views, in that they too are derived data, can speed up queries, and may slow down updates. Thus, the problem of **index selection** is closely related to that of materialized view selection, although it is simpler. We examine index and materialized view selection in more detail in Section 25.1.4.1 and Section 25.1.4.2.

Most database systems provide tools to help the database administrator with index and materialized view selection. These tools examine the history of queries and updates and suggest indices and views to be materialized. The Microsoft SQL Server Database Tuning Assistant, the IBM DB2 Design Advisor, and the Oracle SQL Tuning Wizard are examples of such tools.

16.6 Advanced Topics in Query Optimization

There are a number of opportunities for optimizing queries, beyond those we have seen so far. We examine a few of these in this section.

16.6.1 Top-*K* Optimization

Many queries fetch results sorted on some attributes, and require only the top K results for some K. Sometimes the bound K is specified explicitly. For example, some databases support a **limit** K clause which results in only the top K results being returned by the query. Other databases support alternative ways of specifying similar limits. In other cases, the query may not specify such a limit, but the optimizer may allow a hint to be specified, indicating that only the top K results of the query are likely to be retrieved, even if the query generates more results.

When K is small, a query optimization plan that generates the entire set of results, then sorts and generates the top K, is very inefficient since it discards most of the intermediate results that it computes. Several techniques have been proposed to optimize such *top-K queries*. One approach is to use pipelined plans that can generate the results in sorted order. Another approach is to estimate what is the highest value on the sorted attributes that will appear in the top-K output, and introduce selection predicates that eliminate larger values. If extra tuples beyond the top-K are generated they are discarded, and if too few tuples are generated then the selection condition is changed and the query is re-executed. See the bibliographical notes for references to work on top-K optimization.

16.6.2 Join Minimization

When queries are generated through views, sometimes more relations are joined than are needed for computation of the query. For example, a view v may include the join of *instructor* and *department*, but a use of the view v may use only attributes from *instructor*. The join attribute *dept_name* of *instructor* is a foreign key referencing *department*. Assuming that *instructor.dept_name* has been declared **not null**, the join with *department* can be dropped, with no impact on the query. For under the above assumption, the join with *department* does not eliminate any tuples from *instructor*, nor does it result in extra copies of any *instructor* tuple.

Dropping a relation from a join as above is an example of join minimization. In fact, join minimization can be performed in other situations as well. See the bibliographical notes for references on join minimization.

16.6.3 Optimization of Updates

Update queries often involve subqueries in the **set** and **where** clauses, which must also be taken into account in optimizing the update. Updates that involve a selection on the updated column (e.g., give a 10 percent salary raise to all employees whose salary is $\geq$ \$100,000) must be handled carefully. If the update is done while the selection is being evaluated by an index scan, an updated tuple may be reinserted in the index ahead of the scan and seen again by the scan; the same employee tuple may then get incorrectly updated multiple times (an infinite number of times, in this case). A similar problem also arises with updates involving subqueries whose result is affected by the update.

The problem of an update affecting the execution of a query associated with the update is known as the **Halloween problem** (named so because it was first recognized on a Halloween day, at IBM). The problem can be avoided by executing the queries defining the update first, creating a list of affected tuples, and updating the tuples and indices as the last step. However, breaking up the execution plan in such a fashion increases the execution cost. Update plans can be optimized by checking if the Halloween problem can occur, and if it cannot occur, updates can be performed while the query is being processed, reducing the update overheads. For example, the Halloween problem cannot occur if the update does not affect index attributes. Even if it does, if the updates decrease the value while the index is scanned in increasing order, updated tuples will not be encountered again during the scan. In such cases, the index can be updated even while the query is being executed, reducing the overall cost.

Update queries that result in a large number of updates can also be optimized by collecting the updates as a batch and then applying the batch of updates separately to each affected index. When applying the batch of updates to an index, the batch is first sorted in the index order for that index; such sorting can greatly reduce the amount of random I/O required for updating indices.

Such optimizations of updates are implemented in most database systems. See the bibliographical notes for references to such optimization.

16.6.4 Multiquery Optimization and Shared Scans

When a batch of queries are submitted together, a query optimizer can potentially exploit common subexpressions between the different queries, evaluating them once and reusing them where required. Complex queries may in fact have subexpressions repeated in different parts of the query, which can be similarly exploited to reduce query evaluation cost. Such optimization is known as **multiquery optimization**.

Common subexpression elimination optimizes subexpressions shared by different expressions in a program by computing and storing the result and reusing it wherever the subexpression occurs. Common subexpression elimination is a standard optimization applied on arithmetic expressions by programming-language compilers. Exploiting common subexpressions among evaluation plans chosen for each of a batch of queries is just as useful in database query evaluation, and is implemented by some databases. However, multiquery optimization can do even better in some cases: A query typically has more than one evaluation plan, and a judiciously chosen set of query evaluation plans for the queries may provide for a greater sharing and lesser cost than that afforded by choosing the lowest cost evaluation plan for each query. More details on multiquery optimization may be found in references cited in the bibliographical notes.

Sharing of relation scans between queries is another limited form of multiquery optimization that is implemented in some databases. The **shared-scan** optimization works as follows: Instead of reading the relation repeatedly from disk, once for each query that needs to scan a relation, data are read once from disk, and pipelined to each of

the queries. The shared-scan optimization is particularly useful when multiple queries perform a scan on a single large relation (typically a "fact table").

16.6.5 Parametric Query Optimization

Plan caching, which we saw in Section 16.4.3, is used as a heuristic in many databases. Recall that with plan caching, if a query is invoked with some constants, the plan chosen by the optimizer is cached and reused if the query is submitted again, even if the constants in the query are different. For example, suppose a query takes a department name as a parameter and retrieves all courses of the department. With plan caching, a plan chosen when the query is executed for the first time, say for the Music department, is reused if the query is executed for any other department.

Such reuse of plans by plan caching is reasonable if the optimal query plan is not significantly affected by the exact value of the constants in the query. However, if the plan is affected by the value of the constants, parametric query optimization is an alternative.

In **parametric query optimization**, a query is optimized without being provided specific values for its parameters—for example, *dept_name* in the preceding example. The optimizer then outputs several plans, each optimal for a different parameter value. A plan would be output by the optimizer only if it is optimal for some possible value of the parameters. The set of alternative plans output by the optimizer are stored. When a query is submitted with specific values for its parameters, instead of performing a full optimization, the cheapest plan from the set of alternative plans computed earlier is used. Finding the cheapest such plan usually takes much less time than reoptimization. See the bibliographical notes for references on parametric query optimization.

16.6.6 Adaptive Query Processing

As we noted earlier, query optimization is based on estimates that are at best approximations. Thus, it is possible at times for the optimizer to choose a plan that turns out to perform very badly. Adaptive operators that choose the specific operator at execution time provide a partial solution to this problem. For example, SQL Server supports an adaptive join algorithm that checks the size of its outer input, and chooses either nested loops join, or hash join depending on the size of the outer input.

Many systems also include the ability to monitor the behavior of a plan during query execution, and adapt the plan accordingly. For example, suppose the statistics collected by the system during early stages of the plan's execution (or the execution of subparts of the plan) are found to differ substantially from the optimizers estimates to such an extent that it is clear that the chosen plan is suboptimal. Then an adaptive system may abort the execution, choose a new query execution plan using the statistics collected during the initial execution, and restart execution using the new plan; the statistics collected during the execution of the old plan ensure the old plan is not selected again. Further, the system must avoid repeated aborts and restarts; ideally, the system should ensure that the overall cost of query evaluation is close to that with the

plan that would be chosen if the optimizer had exact statistics. The specific criteria and mechanisms for such adaptive query processing are complex, and are referenced in the bibliographic notes available online.

16.7 Summary

- Given a query, there are generally a variety of methods for computing the answer. It is the responsibility of the system to transform the query as entered by the user into an equivalent query that can be computed more efficiently. The process of finding a good strategy for processing a query is called *query optimization*.
- The evaluation of complex queries involves many accesses to disk. Since the transfer of data from disk is slow relative to the speed of main memory and the CPU of the computer system, it is worthwhile to allocate a considerable amount of processing to choose a method that minimizes disk accesses.
- There are a number of equivalence rules that we can use to transform an expression into an equivalent one. We use these rules to generate systematically all expressions equivalent to the given query.
- Each relational-algebra expression represents a particular sequence of operations. The first step in selecting a query-processing strategy is to find a relational-algebra expression that is equivalent to the given expression and is estimated to cost less to execute.
- The strategy that the database system chooses for evaluating an operation depends on the size of each relation and on the distribution of values within columns. So that they can base the strategy choice on reliable information, database systems may store statistics for each relation r. These statistics include:
  - The number of tuples in the relation r.
  - The size of a record (tuple) of relation r in bytes.
  - The number of distinct values that appear in the relation r for a particular attribute.
- Most database systems use histograms to store the number of values for an attribute within each of several ranges of values. Histograms are often computed using sampling.
- These statistics allow us to estimate the sizes of the results of various operations, as well as the cost of executing the operations. Statistical information about relations is particularly useful when several indices are available to assist in the processing of a query. The presence of these structures has a significant influence on the choice of a query-processing strategy.

- Alternative evaluation plans for each expression can be generated by equivalence rules, and the cheapest plan across all expressions can be chosen. Several optimization techniques are available to reduce the number of alternative expressions and plans that need to be generated.
- We use heuristics to reduce the number of plans considered, and thereby to reduce the cost of optimization. Heuristic rules for transforming relational-algebra queries include "Perform selection operations as early as possible," "Perform projections early," and "Avoid Cartesian products."
- Materialized views can be used to speed up query processing. Incremental view maintenance is needed to efficiently update materialized views when the underlying relations are modified. The differential of an operation can be computed by means of algebraic expressions involving differentials of the inputs of the operation. Other issues related to materialized views include how to optimize queries by making use of available materialized views, and how to select views to be materialized.
- A number of advanced optimization techniques have been proposed, such as top-K optimization, join minimization, optimization of updates, multiquery optimization, and parametric query optimization.

Review Terms

- Query optimization
- Transformation of expressions
- Equivalence of expressions
- Equivalence rules
  - Join commutativity
  - Join associativity
- Minimal set of equivalence rules
- Enumeration of equivalent expressions
- Statistics estimation
- Catalog information
- Size estimation
  - Selection
  - Selectivity
  - Join
- Histograms
- Distinct value estimation
- Random sample
- Choice of evaluation plans
- Interaction of evaluation techniques
- Cost-based optimization
- Join-order optimization
  - Dynamic-programming algorithm
  - Left-deep join order
  - Interesting sort order
- Heuristic optimization
- Plan caching
- Access-plan selection

- Correlated evaluation
- Decorrelation
- Semijoin
- Anti-semijoin
- Materialized views
- Materialized view maintenance
  - Recomputation
  - Incremental maintenance
  - Insertion
  - Deletion
  - Updates
- Query optimization with materialized views
- Index selection
- Materialized view selection
- Top-K optimization
- Join minimization
- Halloween problem
- Multiquery optimization

Practice Exercises

16.1 Download the university database schema and the large university dataset from dbbook.com. Create the university schema on your favorite database, and load the large university dataset. Use the **explain** feature described in Note 16.1 on page 746 to view the plan chosen by the database, in different cases as detailed below.

a. Write a query with an equality condition on *student.name* (which does not have an index), and view the plan chosen.

b. Create an index on the attribute *student.name*, and view the plan chosen for the above query.

c. Create simple queries joining two relations, or three relations, and view the plans chosen.

d. Create a query that computes an aggregate with grouping, and view the plan chosen.

e. Create an SQL query whose chosen plan uses a semijoin operation.

f. Create an SQL query that uses a **not in** clause, with a subquery using aggregation. Observe what plan is chosen.

g. Create a query for which the chosen plan uses correlated evaluation (the way correlated evaluation is represented varies by database, but most databases would show a filter or a project operator with a subplan or subquery).

h. Create an SQL update query that updates a single row in a relation. View the plan chosen for the update query.

i. Create an SQL update query that updates a large number of rows in a relation, using a subquery to compute the new value. View the plan chosen for the update query.

16.2 Show that the following equivalences hold. Explain how you can apply them to improve the efficiency of certain queries:

a. $E_1 \bowtie_\theta (E_2 - E_3) \equiv (E_1 \bowtie_\theta E_2 - E_1 \bowtie_\theta E_3)$.

b. $\sigma_\theta({}_A\gamma_F(E)) \equiv {}_A\gamma_F(\sigma_\theta(E))$, where θ uses only attributes from A.

c. $\sigma_\theta(E_1 \rtimes E_2) \equiv \sigma_\theta(E_1) \rtimes E_2$, where θ uses only attributes from E_1.

16.3 For each of the following pairs of expressions, give instances of relations that show the expressions are not equivalent.

a. $\Pi_A(r - s)$ and $\Pi_A(r) - \Pi_A(s)$.

b. $\sigma_{B<4}({}_A\gamma_{max(B)\ \textbf{as}\ B}(r))$ and ${}_A\gamma_{max(B)\ \textbf{as}\ B}(\sigma_{B<4}(r))$.

c. In the preceding expressions, if both occurrences of *max* were replaced by *min*, would the expressions be equivalent?

d. $(r \bowtie s) \bowtie t$ and $r \bowtie (s \bowtie t)$
In other words, the natural right outer join is not associative.

e. $\sigma_\theta(E_1 \rtimes E_2)$ and $E_1 \rtimes \sigma_\theta(E_2)$, where θ uses only attributes from E_2.

16.4 SQL allows relations with duplicates (Chapter 3), and the multiset version of the relational algebra is defined in Note 3.1 on page 80, Note 3.2 on page 97, and Note 3.3 on page 108. Check which of the equivalence rules 1 through 7.b hold for the multiset version of the relational algebra.

16.5 Consider the relations $r_1(A, B, C)$, $r_2(C, D, E)$, and $r_3(E, F)$, with primary keys A, C, and E, respectively. Assume that r_1 has 1000 tuples, r_2 has 1500 tuples, and r_3 has 750 tuples. Estimate the size of $r_1 \bowtie r_2 \bowtie r_3$, and give an efficient strategy for computing the join.

16.6 Consider the relations $r_1(A, B, C)$, $r_2(C, D, E)$, and $r_3(E, F)$ of Practice Exercise 16.5. Assume that there are no primary keys, except the entire schema. Let $V(C, r_1)$ be 900, $V(C, r_2)$ be 1100, $V(E, r_2)$ be 50, and $V(E, r_3)$ be 100. Assume that r_1 has 1000 tuples, r_2 has 1500 tuples, and r_3 has 750 tuples. Estimate the size of $r_1 \bowtie r_2 \bowtie r_3$ and give an efficient strategy for computing the join.

16.7 Suppose that a B^+-tree index on *building* is available on relation *department* and that no other index is available. What would be the best way to handle the following selections that involve negation?

a. $\sigma_{\neg(building < \text{“Watson”})}(department)$

b. $\sigma_{\neg(building = \text{“Watson”})}(department)$

c. $\sigma_{\neg(building < \text{“Watson”} \vee budget < 50000)}(department)$

16.8 Consider the query:

```
select *
from r, s
where upper(r.A) = upper(s.A);
```

where “upper” is a function that returns its input argument with all lowercase letters replaced by the corresponding uppercase letters.

a. Find out what plan is generated for this query on the database system you use.

b. Some database systems would use a (block) nested-loop join for this query, which can be very inefficient. Briefly explain how hash-join or merge-join can be used for this query.

16.9 Give conditions under which the following expressions are equivalent:

$$_{A,B}\gamma_{agg(C)}(E_1 \bowtie E_2) \quad \text{and} \quad (_{A}\gamma_{agg(C)}(E_1)) \bowtie E_2$$

where *agg* denotes any aggregation operation. How can the above conditions be relaxed if *agg* is one of **min** or **max**?

16.10 Consider the issue of interesting orders in optimization. Suppose you are given a query that computes the natural join of a set of relations S. Given a subset $S1$ of S, what are the interesting orders of $S1$?

16.11 Modify the FindBestPlan(S) function to create a function FindBestPlan(S, O), where O is a desired sort order for S, and which considers interesting sort orders. A *null* order indicates that the order is not relevant. *Hints*: An algorithm A may give the desired order O; if not a sort operation may need to be added to get the desired order. If A is a merge-join, FindBestPlan must be invoked on the two inputs with the desired orders for the inputs.

16.12 Show that, with n relations, there are $(2(n-1))!/(n-1)!$ different join orders. *Hint:* A **complete binary tree** is one where every internal node has exactly two children. Use the fact that the number of different complete binary trees with n leaf nodes is:

$$\frac{1}{n}\binom{2(n-1)}{(n-1)}$$

If you wish, you can derive the formula for the number of complete binary trees with n nodes from the formula for the number of binary trees with n nodes. The number of binary trees with n nodes is:

$$\frac{1}{n+1}\binom{2n}{n}$$

This number is known as the **Catalan number**, and its derivation can be found in any standard textbook on data structures or algorithms.

16.13 Show that the lowest-cost join order can be computed in time $O(3^n)$. Assume that you can store and look up information about a set of relations (such as the optimal join order for the set, and the cost of that join order) in constant time. (If you find this exercise difficult, at least show the looser time bound of $O(2^{2n})$.)

16.14 Show that, if only left-deep join trees are considered, as in the System R optimizer, the time taken to find the most efficient join order is around $n2^n$. Assume that there is only one interesting sort order.

16.15 Consider the bank database of Figure 16.9, where the primary keys are underlined. Construct the following SQL queries for this relational database.

a. Write a nested query on the relation *account* to find, for each branch with name starting with B, all accounts with the maximum balance at the branch.

b. Rewrite the preceding query without using a nested subquery; in other words, decorrelate the query, but in SQL.

c. Give a relational algebra expression using semijoin equivalent to the query.

d. Give a procedure (similar to that described in Section 16.4.4) for decorrelating such queries.

Exercises

16.16 Suppose that a B^+-tree index on (*dept_name*, *building*) is available on relation *department*. What would be the best way to handle the following selection?

$$\sigma_{(building < \text{“Watson”}) \wedge (budget < 55000) \wedge (dept\_name = \text{“Music”})}(department)$$

---

branch(*branch_name*, *branch_city, assets*)
customer (*customer_name*, *customer_street, customer_city*)
loan (*loan_number*, *branch_name, amount*)
borrower (*customer_name*, *loan_number*)
account (*account_number*, *branch_name, balance*)
depositor (*customer_name*, *account_number*)

Figure 16.9 Banking database.

16.17 Show how to derive the following equivalences by a sequence of transformations using the equivalence rules in Section 16.2.1.

a. $\sigma_{\theta_1 \wedge \theta_2 \wedge \theta_3}(E) \equiv \sigma_{\theta_1}(\sigma_{\theta_2}(\sigma_{\theta_3}(E)))$

b. $\sigma_{\theta_1 \wedge \theta_2}(E_1 \bowtie_{\theta_3} E_2) \equiv \sigma_{\theta_1}(E_1 \bowtie_{\theta_3} (\sigma_{\theta_2}(E_2)))$, where θ_2 involves only attributes from E_2

16.18 Consider the two expressions $\sigma_\theta(E_1 \rtimes E_2)$ and $\sigma_\theta(E_1 \bowtie E_2)$.

a. Show using an example that the two expressions are not equivalent in general.

b. Give a simple condition on the predicate θ, which if satisfied will ensure that the two expressions are equivalent.

16.19 A set of equivalence rules is said to be *complete* if, whenever two expressions are equivalent, one can be derived from the other by a sequence of uses of the equivalence rules. Is the set of equivalence rules that we considered in Section 16.2.1 complete? Hint: Consider the equivalence $\sigma_{3=5}(r) \equiv \{\ \}$.

16.20 Explain how to use a histogram to estimate the size of a selection of the form $\sigma_{A \le v}(r)$.

16.21 Suppose two relations r and s have histograms on attributes $r.A$ and $s.A$, respectively, but with different ranges. Suggest how to use the histograms to estimate the size of $r \bowtie s$. Hint: Split the ranges of each histogram further.

16.22 Consider the query

```
select A, B
from   r
where r.B < some (select B
                  from   s
                  where s.A = r.A)
```

Show how to decorrelate this query using the multiset version of the semi join operation.

16.23 Describe how to incrementally maintain the results of the following operations on both insertions and deletions:

a. Union and set difference.

b. Left outer join.

16.24 Give an example of an expression defining a materialized view and two situations (sets of statistics for the input relations and the differentials) such that incremental view maintenance is better than recomputation in one situation, and recomputation is better in the other situation.

16.25 Suppose you want to get answers to $r \bowtie s$ sorted on an attribute of r, and want only the top K answers for some relatively small K. Give a good way of evaluating the query:

a. When the join is on a foreign key of r referencing s, where the foreign key attribute is declared to be not null.

b. When the join is not on a foreign key.

16.26 Consider a relation $r(A, B, C)$, with an index on attribute A. Give an example of a query that can be answered by using the index only, without looking at the tuples in the relation. (Query plans that use only the index, without accessing the actual relation, are called *index-only* plans.)

16.27 Suppose you have an update query U. Give a simple sufficient condition on U that will ensure that the Halloween problem cannot occur, regardless of the execution plan chosen or the indices that exist.

Further Reading

The seminal work of [Selinger et al. (1979)] describes access-path selection in the System R optimizer, which was one of the earliest relational-query optimizers. Query processing in Starburst, described in [Haas et al. (1989)], forms the basis for query optimization in IBM DB2.

[Graefe and McKenna (1993)] describes Volcano, an equivalence-rule–based query optimizer that, along with its successor Cascades ([Graefe (1995)]), forms the basis of query optimization in Microsoft SQL Server. [Moerkotte (2014)] provides extensive textbook coverage of query optimization, including optimizations of the dynamic programming algorithm for join order optimization to avoid considering Cartesian products. Avoiding generation of plans with Cartesian products can result in substantial reduction in optimization cost for common queries.

The bibliographic notes for this chapter, available online, provides references to research on a variety of optimization techniques, including optimization of queries with aggregates, with outer joins, nested subqueries, top-K queries, join minimization, optimization of update queries, materialized view maintenance and view matching, index and materialized view selection, parametric query optimization, and multiquery optimization.

Bibliography

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